Order Marsupialia

The taxonomically and ecologically diverse marsupials give birth to extremely altricial neonates with perinatal specializations of the pectoral girdle and digits (5863). Lactation is long and biphasic while gestation is short. Neontates are an exceptionally small proportion of maternal mass and embryogenesis is completed during the first phase of lactation, when neonates are continuously attached to a teat. Marsupial development at teat detachment (first pouch exit for those marsupials with pouches) is often comparable to that of eutherian neonates at birth. A pouch encloses the mammary teats in many, but not all, families. Pouches are not an ancestral trait for marsupials and probably have evolved independently several times. Females have a duplex uterus with two cervices and a median birth canal between the lateral vaginae. The birth canal in some families is reformed with each pregnancy. Only peramelids use chorioallantoic placentation, all other marsupials use choriovitelline placentation only. Testicular vasculature includes a rete mirabile (4136). The prostate is disseminate. The penis is often bifid except in Notoryctidae and Tarsipidae and lacks a baculum. When present, the scrotum is prepenial. The spermatazoan acrosome is exceptionally stable (6941).

Unlike eutherians, X-chromosome inactivation in marsupials is incomplete and non-random (the paternal X is inactivated; 4013). Scrotal development occurs when a single X-chromosome is present (even in the absence of a Y-chromosome), while pouch development proceeds with a pair of X-chromosomes (even in the presence of a Y-chromosome; 4013).

Reviews exist on the following topics: reproductive physiology (9852, 11008, 10998a), captive breeding (1987, 3721, 9205), anatomy (8292, 11098, 11100), sperm (2169), sperm pairing (934), sex differentiation (9041a), placentation (5075, 9038), parental investment (9378, 9379), and milk composition (4026). More general reviews exist for several taxonomic or ecological groups: Australian mammals (10474); carnivorous marsupials (356); possums and opossums (357); bandicoots and bilbies (9719); possums and gliders (10088); kangaroos, wallabies, and rat-kangaroos (4071).

Order Marsupialia, Family Didelphidae

New World opossums (*Caluromys* 130-500 g; *Caluromysiops* 250-300 mm; *Chironectes* 600-800 g; *Didelphis* 1-5.5 kg; *Glironia* 160-205 mm; *Lestodelphys* 132-144 mm; *Lutreolina* 200-540 g; *Marmosa* 10-150 g; *Metachirus* 400-800 g; *Monodelphis* 20-155 g; *Philander* 222-1500 g) are nocturnal, terrestrial to semiarboreal (*Chironectes* is semiaquatic) omnivores of forested areas primarily in Central and South America (*Didelphis* is found in North America). Data on *Caluromysiops*, *Glironia*, and *Lestodelphys* are limited.

Didelphid males are often larger than females (439, 3024, 4976, 8555, 11007; cf 5210). Copulation can last over 2 h (*Marmosa* 604, 606, 2956). As large numbers of eggs may be ovulated (3123, 4443, 8850), litter size is limited by the number of mammae: 4 (*Glironia* 6859), 4-5 (*Chironectes* 6860), 4-13 (*Marmosa* 2956, 3024, 3840, 6022), 5-9 (*Philander* 7983), 5-11 (*Metachirus* 3024, 6021), 6-13 (*Monodelphis* 8552, 8555), 7-17 (*Didelphis* 630, 3024, 4284, 4437, 4438, 5146, 6022, 7097, 7497, 9074, 9745), and 9 mammae (*Lutreolina* 6857).

Neonates open their eyes at 2-5 wk (*Monodelphis* 6009, 6744), 35-40 d (*Marmosa* 608, 2937, 2956), or 50-72 d (*Didelphis* 4444, 9074). *Caluromys*, *Caluromysiops*, *Didelphis*, and *Philander* females have pouches, while *Glironia*, *Lutreolina*, *Marmosa*, *Metachirus*, and *Monodelphis* do not. Both male and female *Chironectes* have a posteriorly opening pouch (3024, 9292). Most didelphids are polyestrous, although *Didelphis albiventris* may be monoestrous (10494) and *Monodelphis dimidiata* may be semelparous (8555). *Marmosa* (3840) and *Monodelphis* (3139) have bimodally distributed estrous cycle lengths. The shorter cycles (about equal to the gestation length) are sterile. At 2 wk the length of gestation is about half the usual estrous cycle length, both of which are relatively constant across the family. Lactation is much longer (1.5-3 mo) and more variable. Earliest sexual maturity is at about twice the lactation length (3-6 mo). Polyovular follicles are common for *Didelphis* (4443).

Didelphid sperm become paired in the epididymides (934a). The bifid penis is posterior to the scrota in *Philander* and *Caluromys* (1423, 9088a).

Details of placentation (6009b, 6010, 6011), shell (6009a), anatomy (484, 490, 604, 3315, 3352, 3845, 4439, 4766, 5201, 6154, 6916, 7830, 7948, 7948a, 8484, 9204, 9746, 11012), endocrinology (2032, 3108a, 3138, 3722, 4363, 4364, 4365, 6155, 6916, 8083a, 8380, 11309, 12093), sex differentiation (1463a, 1463b, 1463c, 1463e, 3141, 4436, 7491c, 7491d, 7507b, 9045a, 9355a, 9355b), ovulation/conception (606, 710, 3136, 3137, 3139, 3840, 4439, 6916, 9206, 9207), spermatogenesis (710, 934, 6694, 8129), and milk composition (807a, 2143) are available.

NEONATAL MASS		
Caluromys philander	200 mg	438, 439
Didelphis marsupialis	~60 mg	630
	150 mg	4452
	160 mg	10994
	200 mg	439
	~390 mg	5146
Didelphis virginiana	130 mg	4444
	150 mg; r140-190; n4	7492
	160 mg	4284
Marmosa murina	90 mg	2956
Marmosa robinsoni	~100 mg	608
Monodelphis dimidiata	70-80 mg	2222
	r80-110 mg; n3	8555
Monodelphis domestica	100 mg	3142, 11215
Philander opossum	200 mg	439
NEONATAL SIZE		
Caluromys philander	10 mm	438, 439
Didelphis marsupialis	10 mm	439
	~10 mm	5146
Didelphis virginiana	HB: 13.8 mm; r13.0-15.4; n28 yg, 9 lt, 1 d	9074
	CR: 14 mm	4284
	14 mm	7492
Marmosa cinerea	10 mm	439
Marmosa murina	10 mm	439
Marmosa robinsoni	6-14 mm	451
	8-12 mm	1987
Monodelphis dimidiata	CR: ~10 mm	8555
Monodelphis domestica	10 mm	11215
Philander opossum	10 mm	439
WEANING MASS		
Caluromys philander	11 g; 75-80 d pouch exit	438
Didelphis spp	75-105 g	11007
Didelphis marsupialis	25 g; controls; 39 g; food supplemented	461
Didelphis virginiana	120-150 g	1464
	150 g	5473
Marmosa robinsoni	10-12 g	608
Monodelphis domestica	3-5 g; 3 wk	6744
	r16.53-24.05 g; n4 groups, n35 lt, 56 d	2063
Philander opossum	100-120 g	1987
WEANING SIZE		
Caluromys philander	75 mm; 75-80 d pouch exit	438
Didelphis virginiana	HB: 195.2 mm; r171.6-217.7; n41 yg, 10 lt, 95 d	9074
LITTER SIZE		
Caluromys spp	mode 3-4; r1-6	3027
Caluromys derbianus	2 emb; n1 f	931
	2.75 yg; r2-3; n4 f	931
	3 yg; n1 lt	4228
	3.3; r2-4; n13 lt	8482
	4 yg; n1 f	5423c
Caluromys philander	4 yg; n1 lt	4766
	4.1 yg; r2-7; n42 lt	439
	4.2 yg; r2-7; n28	438
	6 yg ea; n3 lt	7994
Caluromysiops irrupta	2 yg; n1 f; via photo	5201

Chironectes minimus	2-3	3027
	4.67; r4-5; n3 lt	9292
	max 5	7465
Didelphis albiventris	4.2; sd1.40; mode 4-5; r2-7; n10 lt	11007
	6.2; sd2.2; r3-9; n12 lt	10494
	6.5 emb; ±2.4; r3-10; n35 f	1675
Didelphis aurita	19-31 oocytes	4756
Didelphis marsupialis	4-6 yg	5111
	4.49; sd1.36; mode 5; r1-7; n41 lt	11007
	4.5; r2-7; n42 lt	439
	4.67 yg; r2-7; n3 f	3022, 3024
	5-10 yg	5146
	5 yg; n1 f	11751
	5.9; r2-10; n15	7994, 7998
	6.0 yg; r2-9; n29	3320
	6.54; sd2.19; mode 7; r1-11; n37 lt	11007
	7 yg; n1 lt	9029
	7.5; r3-10	7994, 7995, 7998
	7.5; r6-9; n2 lt	12128
	7.52; se0.36; n18 f	461
	7.61; se0.34; n20 f	461
	8 yg; n1 lt, <7 wk	2064
	mode 8-9; r7-10	6021
	9.78 yg; r5-17; n9 f	7624
	10 yg; n1 f	6084a
	11.7; r11-12; n3	7576
Didelphis virginiana	6 yg; n1 f	5423c
	6.26; mode 7; r3-11; n50 lt	1464, 1465
	6.7; r1-14	7492
	6.8 yg; n65 lt	6295
	6.8 yg; r1-10; n68 lt	2899
	7 yg ea; n2 f	1290
	7.1 yg; se0.18; r1-11; n143 f	7077
	7.2 yg; mode 7; r4-11; n44 lt	9074
	7.33 yg; r6-9; n3 f	10471
	7.4 yg; mode 7; r1-12; n28 lt	3286
	7.74 yg; n57 lt; 8.5 yg ≤40 mm; 6.6 yg >40 mm	6514
	7.9 yg; n85 lt	4930
	7.91; min 1; n58 lt	9465
	8 yg; n1 f	2191
	8.0 yg; r6-9; n4 lt	3212
	8.2 yg; r5-11; n5 lt; not at birth	4976
	8.6; n23 lt	9074
	8.65; ±1.83; mode 8-10; r6-12; n31 lt	2190
	8.66; mode 9; r2-15; n346 lt	4284
	8.8 yg; r3-13; n10 lt	6326
	8.9 yg; mode 8; r5-13; n42 lt	9073
	9 yg; r6-12; n7 lt	11834
	9.67 yg; r7-13; n3 f	8907
	mode 12-16 emb; r7-27	9745
	13 yg; n1 f	970
	14 yg; n1 f	1290
	19.5 yg; r18-21; n2 f; max 13-14 attach	4438, 4446
	22 ova; n1	10283
	22 ova; max 43	4438
	23 emb; n1 f	4446
	max 25 emb	4284
	29.6 oocytes; r19-40	9206
	r30-56 ova, blastocysts; n11 f	4446
	max 50 ovulations	1464
	max 70 ovulations	4444
Lutreolina crassicaudata	7 yg; n1 lt	6372
	mode 8-9; r7-10	6021

Marmosa spp	2; n1 lt	3021
	6 yg; n1 f	5063
	9; n1 lt	8692
	9.0; r8-10; n2 lt	2937
Marmosa canescens	8 yg ea; n2 f	5410
	13 emb; n1 f	5410
Marmosa cinerea	2 yg; n1	10683
	5 yg; n1 f	676
	6.2; max 11; n4 lt	439
Marmosa constantiae	8 yg; r7-9; n2	10683
Marmosa elegans	17 emb; n1 lt	6022
Marmosa fuscata	6	7995, 7998
Marmosa lepida	5 yg; n1 f	10683
Marmosa marica	6 yg	10683
Marmosa mexicana	10 yg; n1 f	1475
	11 yg; n1 lt	3750
	13 yg; n1 f	3212
Marmosa murina	5.5 yg; r4-7; n2 f	5111
	5-6 yg	5111
	6 yg	5111
	8.4; max 11; n33 lt	439
	12; r11-13; n3 lt	2956
Marmosa parvidens	6.5 emb; r6-7; n2 f	8553
Marmosa robinsoni	4.8-7.07 yg; n3 samples	604
	6.0; n70 lt	606
	6.91 yg; r1-12; n11 lt	10813
	7.8; sd3.1; n18 lt parental generation; 8.2; sd3.6; n15 lt F_1; 9.1; sd2.7; n16 lt F_2; r1-13	3840
	9.0; r8-10; n2 f	2937
	10.0 yg; r6-13; n16	3320
	12 yg; n1 lt	605
	12.25 yg; r3-16; n4 f; not birth	451
	14.0; r13-15; n13	7995, 7998
	19.8 cl; ±0.9; mode 21; n23	3840
Marmosa tyleriana	3 enlarged teats; n1	10683
Metachirus nudicaudatus	2 yg; r1-3; n2 lt	3024
	max 9	1987
Monodelphis brevicaudata	6 yg	5111
	7.5; r7-8; n5 lt	7995, 7998
Monodelphis dimidiata	12; r8-16; n2 lt	2222, 8555
Monodelphis domestica	1-11	10493
	3.79-5.55 wean	2063
	5-13; 13 rare	11215
	6 ovulations/ovary	490
	6.00-7.45 yg	2063
	7 yg; r3-14	3142
	8	10797
	8.4; sd2.6; r6-11	10494
	8.6 yg; sd2.6; r2-13	3136, 6009
Philander spp	1	9746
Philander opossum	3.3; n11, 3 lt	3024
	3.43 yg; r1-7; n7 f	5111
	3.67 yg; r2-5; n3 lt	3024
	4.2; r2-7; n179 lt	439
	4.6 yg; r2-7; n34	3320
	4.75 yg; r4-5; n4 lt	3212
	5.0 yg; r4-6; n2 lt	4228
	5.75 yg; r4-7; n4 f	7624
	6 yg; n1 f	5423c
	6.0 yg; r5-7; n2 f	2341
	6.05; mode 7; r3-7; n21 lt	8483
	7 yg; n1 lt	4766
	8 emb; n1 lt	3016

SEXUAL MATURITY

Caluromys derbianus	f/m: 7-9 mo	1987
Caluromys philander	f: 1st lact: 10 mo	439
Chironectes minimus	f: 1st estrus: ~10 mo	8093
Didelphis spp	f/m: 7 mo	11007
Didelphis albiventris	f/m: ~1 yr	10494
Didelphis marsupialis	f: 1st mate: 6 mo	5146
	f: 1st mate: 800 g	4452
	f: 1st lact: 6 mo	439
	m: 1st mate: 8 mo	5146
	f/m: 6 mo; n26	7998
Didelphis virginiana	f: 1st birth: 186 d	9074
	f: 6 mo	9074
	m: testes descend: 77 d	3256
	m: glands mature: 100 d	3256
	m: 8 mo	9074
Marmosa fuscata	f/m: 6 mo; n10	7998
Marmosa robinsoni	f: 1st estrus: 40 wk; 40 g; n200	608
	f: 1st mate: 278 d	604
	m: 1st mate: 1 yr	608
	f/m: 6 mo; n13	7998
Monodelphis brevicaudata	f/m: 6 mo; n5	7998
Monodelphis domestica	f/m: 3-5 mo	10921
	f/m: 3.5 mo	6744
	f/m: 4-5 mo	11215
	f/m: 5-7 mo	10494
Philander opossum	f: 1st estrus: 15 mo	1987
	f: 1st lact: 8 mo	439

CYCLE LENGTH

Caluromys derbianus	mode 27-29 d; r16-39; n1 f	1422
Caluromys lanatus	mode 27-29 d; r20-31; n3 f	1422
Caluromys philander	38.6 d; se1.4; r28-45; n14 cycles, 4 f	8380, 8380a
Chironectes minimus	estrus: max 24 h	8093
Didelphis marsupialis	7 d	6915
Didelphis virginiana	estrus: 3-5 h	9745
	estrus: 36 h	9074
	23-28 d	3174
	25.5 d; r17-38	5473
	28 d; r22-45	4439
	4-6 wk	9745
	29.2 d; se1.0; r22-42; n24 cycles, 18 f	3315
	29.5 d; r22-38; n8-10 f; increases throughout season	9074
	~1 mo	4438
Marmosa robinsoni	estrus: ~3 d	3840
	15.6 d; se0.7; mode 15; r13-19; n10 cycles, 36 f	3840
	23 d; r18-31	604
	25.5 d; se0.5; mode 24; r23-30; n23 cycles, 21 f	3840
	27 d	605
	mode 28 d; n500 observations	608
Monodelphis domestica	estrus: 5.9 d; sd2.11; r3-12	3139
	estrus: r4-8 d; n6 f; 12 d; n1 f	3136
	estrus occurred: 8.5 d after m introduced; 95%ci 7.56-9.2; r8.5-28; n80 f;	490
	14.4; sd2.8; r11-17; n5	3139
	28 d	11215
	~1 mo	10921
	~30 d	3140
	32.3; sd3.4; r28-39; n10	3139

GESTATION		
Caluromys philander	24 d; se0.9; n13	8380a
Didelphis spp	13 d	2937
Didelphis albiventris	12 d; n1	11007
Didelphis marsupialis	12-13 d	5146
	13 d	6916
	2 wk	6021
Didelphis virginiana	11 d	4438
	12.6 d; n1	4439
	12 d, 18+ h	1465
	12.8 d	9745
	12-13 d	7492
	12-13 d	4439, 4444
	13 d	4284
	13 d	4452
	13 d; n1	3594
	13 d; r12.85-13.19; n18 f	9074
Lutreolina crassicaudata	2 wk	6021
Marmosa spp	14 d	2937
Marmosa murina	≤13 d; last paired to birth	2956
Marmosa robinsoni	min 13 d; pairing to birth	10813
	min 14 d; pairing to birth	605
	13.5-14.5 d	3840
	14 d	2937
	14 d	608
	~15 d	604
Monodelphis domestica	14 d	10797
	14-15 d	3136, 3140, 3142, 11215
	15 d	10921
LACTATION		
Caluromys philander	pouch life: 75-80 d	438, 439
	wean: ~100 d (4-6 wk after teat detachment)	438
Chironectes minimus	teat attachment: 48 d	9292
Didelphis spp	wean: ~100 d	11007
Didelphis marsupialis	pouch life: 80 d	439
	wean: ~60 d	461
	wean: 90-100 d	5146
Didelphis virginiana	teat attachment: 50-65 d	7097
	solid food: 75 d	11834
	solid food: 85 d	2190
	solid food: 92 d	9074
	pouch exit: 80-87 d	9074
	pouch exit: 90-100 d	5473
	wean: 80-90 d	4444
	wean: 96-104 d	9074
	wean: 4 mo	1464
	forage: 90 d	11834
Marmosa spp	teat attachment: 20 d	2937
Marmosa murina	wean: 62 d	2956
Marmosa robinsoni	teat attachment: 20 d	2937
	teat attachment: 21 d	608
	solid food: 50-58 d	1987
	wean: 60-70 d	604, 608
	wean: 65-75 d	10814
	wean: 70 d	605
Monodelphis dimidiata	teat attachment: <2 wk	8555
Monodelphis domestica	teat attachment: <14 d	4522
	teat attachment: 14-17 d	6009
	teat attachment: 3 wk	6744
	den emergence: 4 wk	6744
	solid food: 4-5 wk	3142
	nurse: 6-8 wk	10494
	wean: 49-56 d	6009
	wean: 50 d	11215

	independent: 7 wk	3142
	wean: 56 d	10797
Philander opossum	pouch life: 80 d	439
INTERLITTER INTERVAL		
Didelphis virginiana	4 mo	9074
Monodelphis domestica	min 7-8 wk	10494
LITTERS/YEAR		
Caluromys philander	2	439
Didelphis albiventris	1	10493, 10494
Didelphis marsupialis	1.4	7998
	2	5146
	2 4	39
	est 2	3320
	est 2+	630
Didelphis virginiana	1	9745
	1-2	4930
	1-2	9074
	1-3	7492
	2	6295
	2	4284
	2	3286
	2	5473
	2; r1-3?; no good evidence for 3	4439, 4444
Marmosa fuscata	1.6	7998
Marmosa robinsoni	3.6	7998
Monodelphis brevicaudata	1.5	7998
Monodelphis dimidiata	1	8555
Monodelphis domestica	max 4	11215
	max est 5-6	10494
Philander opossum	3	439
SEASONALITY		
Caluromys derbianus	repro: est yr rd; Nicaragua	931, 8482
Caluromys philander	preg f: Apr-June, Oct-Dec; French Guyana	438
	lact f: low July-Sept; French Guyana	8380
Chironectes minimus	birth: Dec-Jan; Brazil	1519
Didelphis albiventris	m fertile: yr rd; ne Brazil	1675
	mate: Mar-Sept; Colombia	11007
	birth: Nov-Dec; ne Brazil	10494
	birth: peak Nov-Dec; ne Brazil	6808
Didelphis aurita	preg/lact f: June-July, Oct; Brazil	4756
Didelphis marsupialis	sperm: yr rd; Panama	3320
	sperm: yr rd; Nicaragua	930
	anestrus: Oct-Dec; Panama	3320
	mate: Jan-Aug; e Colombia	11007
	mate: Apr-Sept; not sampled later; w Colombia	11007
	mate: Jan-Sept; Panama	3320
	mate: Oct-Nov, again when yg independent; Argentina	6021
	mate: peak end Dec; Marago I, Brazil	2064
	conceive: peaks Jan-early June, Aug; Nicaragua	930
	lact f: Mar, May, Aug; n3; Barro Colorado I, Panama	3024
	lact f: Mar-Oct, n Venezuela	7994
Didelphis virginiana	sperm: yr rd; TX	4444
	anestrus: Oct-Dec, lab, PA	3174
	uteri large, cl: Feb-July; DC	7508
	estrus: mid Jan-mid Mar; FL	1465
	estrus: end Feb-mid Apr; captive	9745
	mate: Jan-?; lab, PA	3174
	mate: peak mid Jan-Apr, none Nov-Dec; TX	4439, 4444
	mate: peak Feb-Mar; NE	5405b,
	ovulate: Dec-July; captive	5473
	conceive: mid Jan-June; central CA	9074
	conceive: Oct-July, peaks Jan-Feb, mid Apr-early May, Oct; OR	4976

	birth: Jan; FL, LA, AK	7492
	birth: Feb; MI	7492
	birth: end Feb-mid June; IL	7492
	birth: Jan-July; captive	5473
	birth: end Jan-Mar, peak Feb 31/33; FL	1464
	birth: end Jan-mid Mar, peak end Jan-mid Feb; n47 lt; FL	1465
	birth: Feb-June; MO	9073
	birth: end Feb-early July; MD	6514
	birth: Mar-July; KS	3286
	lact f: Jan-Sept, none Oct-Dec; TX	4438, 4444
	lact f: end Jan-mid Oct; central CA	9074
	lact f: Feb-?; IA	11834
	lact f: Feb-Mar, Aug; n3; NC	1290
	lact f: mid Feb-July; GA, FL	7077
	lact f: Mar-June; El Salvador	3212
	lact f: Aug; n1; Cozumel I, Qunitana Roo, Mexico	5423c
Lutreolina crassicaudata	mate: Oct-Nov, again when yg independent; Argentina	6021
Marmosa alstoni	preg/lact f: Aug, Oct; n3; Mexico	10683
Marmosa cinerea	preg/lact f: Jan-Mar, Sept-Nov; n10; n S America	10683
Marmosa constantiae	preg/lact f: Jan; n2; Bolivia	10683
Marmosa elegans	preg/lact f: Feb, Sept-Oct; n3; Chile, Bolivia	10683
Marmosa fuscata	lact f: May-Nov; few data; n Venezuela	7994
Marmosa germana	preg/lact f: Jan, Mar-Apr, Sept-Oct; n9; Ecuador, Peru	10683
Marmosa grisea	preg/lact f: Mar; n1; Bolivia	10683
Marmosa juninensis	preg/lact f: Nov; n1; Peru	10683
Marmosa lepida	preg/lact f: Jan, Mar; n2; Ecuador, Bolivia	10683
Marmosa marica	preg/lact f: Feb, July; n2; Venezuela	10683
Marmosa mexicana	lact f: June; El Salvador	3212
Marmosa murina	preg/lact f: Feb-June, Aug, Oct; n10; n S America	10683
Marmosa noctivaga	preg/lact f: Jan, Apr, Oct; n4; nw S America	10683
Marmosa parvidens	preg f: June-July; n2; Brazil	8553
Marmosa phaea	preg/lact f: May; n1; Colombia, Ecuador	10683
Marmosa quichua	preg/lact f: July, Oct; n3; Peru	10683
Marmosa rapposa	preg/lact f: Nov-Dec; n3; Peru	10683
Marmosa robinsoni	sperm: yr rd; Panama	3320
	anestrus: Oct-Feb; Panama	3320
	mate: Mar-Sept; Panama	3320
	preg f: none June, Aug, Nov-Apr; n40 f; Barro Colorado I, Panama	3024
	preg/lact f: Apr-June; Venezuela	451
	preg/lact f: May-Sept, Dec; n28; s Central America, n S America	10683
	lact f: May-Nov; few data; n Venezuela	7994
Marmosa rubra	preg/lact f: Mar-Apr, Dec; n4; Ecuador	10683
Marmosa tyleriana	preg/lact f: Feb; n1; s Venezuela	10683
Monodelphis brevicaudata	lact f: May-Aug, Nov; few data; n Venezuela	7994
Monodelphis dimidiata	birth: Dec-Jan; Argentina	8555
Monodelphis domestica	mate: yr rd; lab	10921
	mate: yr rd; lab	6744
	birth: yr rd; lab	11215
	lact f: min 8 mo of yr; ne Brazil	10493, 10494
	repro: aseasonal; ne Brazil	6808
Philander opossum	sperm: est yr rd; Panama	3320
	anestrus: Oct-Dec; Panama	3320
	mate: Jan-Sept; Panama	3320
	lact f: Feb-Oct; Nicaragua	930
	lact f: Mar; n1; Campeche, Mexico	5423c
	lact f: Mar-July; Nicaragua	8483
	lact f: Mar-Aug; El Salvador	3212
	lact f: Apr-July; Barro Colorado I, Panama	3024
	lact f: June, Sept-Oct; Venezuela	1987
	repro: est not seasonal; Veracruz, Mexico	4228

Order Marsupialia, Family Microbiotheriidae, Genus *Dromiciops*

The 20-40 g, Chilian microbiothere, *Dromiciops australis*, is a nocturnal, scansorial inhabitant of cool, moist forests where it may live in pairs (6773, 6777, 8556). Arthropods predominate in its diet. Males have scrotal testes and females have 4 teats in a small pouch (6777). Both scrotum and pouch are covered with rufous hair. Few accounts of its biology exist (3265, 6858).

LITTER SIZE		
Dromiciops australis	1 yg; n1 f	8305a
	2 yg; n1	6777
	2 yg; n1	4054
	3 yg; n1	8138
	4 yg; n1	6022
	5 yg; n1 nest	8478
SEXUAL MATURITY		
Dromiciops australis	f/m: 2nd yr	6777
SEASONALITY		
Dromiciops australis	lact f: Nov; n1; Chile	8138
	lact f: Dec; n1; Patagonia, Argentina	8305a
	lact f: Dec; n1; Malleco, Chile	4054
	repro: spring; Chile	6858

Order Marsupialia, Family Caenolestidae

The 15-45 g caenolestids (*Caenolestes*, *Lestoros*, *Rhyncholestes*) are terrestrial, nocturnal insectivores of South American cold and wet forests (5821, 6861). Females have 4 (*Caenolestes*, *Lestoros*) to 7 (*Rhyncholestes*) mammae, but no pouch (5821, 8257). *Caenolestes* males have a prostate gland 6 times the volume of their testes, 3 pairs of bulbourethral glands, a deeply forked glans penis (females have a deeply bifid clitoris; 8136, 9204). Distinctive sperm become paired in the epididymides (934a). Details of caenolestid anatomy are available (672, 9204). The reported reproductive biology of caenolestids is extremely limited.

LITTER SIZE		
Caenolestes spp	3 emb; n1 f	8136
Caenolestes caniventer	2 emb; n1 f	609
Caenolestes obscurus	3-4 enlarged teats; n4 f	5821
SEASONALITY		
Caenolestes obscurus	lact f: Aug; s Colombia	5821
Rhyncholestes raphanurus	repro f: Dec; n1; Chile	7242

Order Marsupialia, Family Dasyuridae

Dasyurids (*Antechinus* 10-120 g; *Dasycercus* 60-175 g; *Dasyuroides* 70-170 g; *Dasyurus* 0.4-4 kg; *Murexia* 105-285 mm; *Myoictis* 170-250 mm; *Neophascogale* 170-230 mm; *Ningaui* 8-13 g; *Phascogale* 45-200 g; *Phascolosorex* 117-226 mm; *Planigale* 5-15 g; *Sarcophilus* 4-18 kg; *Sminthopsis* 10-50 g) are terrestrial, nocturnal carnivores and insectivores from Australia and New Guinea. The New Guinean genera *Murexia*, *Myoictis*, *Neophascogale*, and *Phascolosorex* are poorly known.

Dasyurid males are often larger than females (423, 732, 1540, 2543, 2837, 3304, 3307, 4989, 5077, 6344, 7552, 7750, 9595, 10699, 11646, 11914; cf 1540, 4989). Copulation may be lengthy, e.g., 5-12 h in *Antechinus* (1194, 3131, 3163, 3305, 3439, 3601, 6344, 6825, 7280, 10642, 11646, 11913, 11915, 11922a; cf 423).

Sperm storage (710, 9757, 9761) and delayed development (9755) occur in *Antechinus*. Estrus has not been recorded post partum (425, 10126). In some species (e.g., *Antechinus*, *Dasyurus*, *Phascogale*) males die immediately after the mating season (589b, 609a, 731, 1190, 1191, 1192a, 1193, 1194, 1195, 1209, 1768a, 1936, 2174, 2189, 2543, 2553a, 4029a, 6307a, 7046a, 7047, 7455, 9699a, 10698, 11392, 11748, 11886, 11910b, 11913, 11917b, 11922a).

Pouch young open their eyes at 34-43 d (*Planigale* 424, 11646), 48-50 d (*Ningaui* 3163; *Sminthopsis* 3131, 3439, 3843; cf 10642), 58-70 d (*Antechinus* 6825, 9804, 11721, 11746, 11912), 60-74 d (*Dasycercus* 3305,

7280), 67-78 d (*Dasyuroides* 423, 425, 3664, 7181), 72-91 d (*Dasyurus* 2028, 3297, 7221, 9775; cf 3301, 9803), 87-105 d (*Sarcophilus* 3298, 3304, 4132).

Females have 4 (*Sarcophilus* 3304, 4040, 4132, 5076, 5077; *Phascolosorex* 7983; *Murexia* 7983), 4-12 (*Antechinus* 610, 731, 1540, 1935, 1936, 2554, 2838, 2841, 4043, 9757, 10108, 10703, 11392, 11395, 11396, 11425, 11721, 11746, 11748, 11886), 5-7 (*Dasyuroides* 423, 425, 11914), 5-15 (*Planigale* 95, 7983, 8992, 10699, 11646), 6 (*Myoictis*, 7983), 6-7 (*Ningaui* 354, 3163), 6-8 (*Dasycercus* 3305, 10698; *Dasyurus* 732, 3297, 4040, 11511, 11512), 6-11 (*Sminthopsis* 3838, 3843, 5836, 6436, 7552, 11923, 11927), or 8-13 mammae (*Phascogale* 2189).

Dasyurid marsupia vary from relatively undeveloped, flat, or non-existent (*Antechinus*, *Dasycercus*, *Dasyuroides*, *Dasyurus*, *Myoictis*, *Phascogale*) to extensive, deep pouches (*Antechinomys*, *Planigale*, *Sarcophilus*, *Sminthopsis*; 2189, 3163, 3304, 3439, 4989, 8617, 9852, 11916). An accessory erectile structure ventral to the urethral penis is present in some species (353, 11917a).

Details of anatomy (3351, 3352, 4754, 5077, 8292, 8297, 9761a, 11098, 11100, 11903, 11917), endocrinology (1388, 3326, 3327, 4803, 4808, 9699a, 9790, 11747, 11748, 11921, 11922a), sex differentiation (11029), ovulation/conception (1274, 4755, 4773, 5076, 7999, 9470, 9756, 9758, 9759, 9760, 10129), spermatozoa (10619), placentation (4754, 5075, 5076a), birth (5116), milk composition (4026, 4029, 7246), and light (2553b, 3839, 7006, 10126) are available.

NEONATAL MASS

Antechinus macdonnellensis	12.3 mg; n1	11922
Antechinus stuartii	15 mg	11912
	16.4 mg	6825
Dasyuroides bymei	26.8 mg; sd4.3; n62	3326
Dasyurus geoffroii	11 mg; r9-13; n2 in alcohol	9775
	14.5 mg; n1	376
Dasyurus viverrinus	20 mg	7218
Ningaui spp	5 mg	3163
Planigale gilesi	9.7 mg; n1	11646
Sarcophilus harrisii	18-29 mg	4132
Sminthopsis macroura	10 mg; n1	3838

NEONATAL SIZE

Antechinus bilami	CR: 7.2 mm; sd1.3	731
Antechinus macdonnellensis	CR: 3.8 mm; n1	11922
Antechinus minimus	3-5 mm	11746
Antechinus rosamondae	CR: 3.0-3.5 mm	11922a
Antechinus stuartii	CR: 4.9 mm	6825
Antechinus swainsonii	4.6 mm; r4.5-4.7	11721
Dasyuroides bymei	4 mm	423
	4 mm	6697
	5.82 mm; sd0.37; n62	3326
Dasyurus geoffroii	4.4 mm; n2	9775
Dasyurus viverrinus	4.7-4.9 mm	1387
	7.6 mm	4754
Ningaui spp	3.5-4.5 mm	3163
Planigale gilesi	2.0 mm	11646
Planigale maculata	f: 5.7 mm	10699
Sarcophilus harrisii	12.5 mm; n4	3304
Sminthopsis crassicaudata	CR: 3.5-4 mm; from graph	7552
Sminthopsis longicaudata	CR: 3.0 mm	11927
Sminthopsis macroura	f: CR: 4.1 mm; n1	3838
Sminthopsis murina	5.4 mm; 3 d	3439

WEANING MASS

Antechinus rosamondae	8.2 g; n1. 90 d	11922a
Dasycercus cristicauda	~22 g; 100-110 d	7280
Dasyuroides bymei	~45 g	423
Dasyurus geoffroii	f: 35.2 g; sd2.0; n8, 72 d	9775
	m: 37.0 g; sd2.0; n5, 72 d	9775
Dasyurus viverrinus	f: 379.2 g; n19, 135 d	7221
	400 g	7218
	m: 465.8 g; n26, 135 d	7221

	f: 472.1 g; n17, 155 d	7221
	m: 652.9 g; n19, 155 d	7221
	f: 713.8 g; n10, 205 d	7221
	m: 1131.4 g; n11, 205 d	7221
Ningaui spp	2.5 g	3163
Planigale gilesi	f: 5.81 g; n5; independence	6519
	m: 6.03 g; n3; independence	6519
Planigale maculata	4.6-5.1 g	424
Planigale tenuirostris	f: 2.67 g; n4; independence	6519
	m: 2.93 g; n6; independence	6519
Sarcophilus harrisii	~200 g	4132
Sminthopsis crassicaudata	5-8 g	3843
Sminthopsis leucopus	f: 9 g	11923
	m: 11 g	11923
Sminthopsis macroura	7 g; r5.3-8.2	3838
Sminthopsis virginiae	18-20 g	10642
WEANING SIZE		
Ningaui spp	21.5 mm	3163
LITTER SIZE		
Antechinus apicalis	7; n1 lt	10917
	8; n1 lt	11915
Antechinus bellus	est 12+ ova	10703
	14 emb; r11-16	1540
	15 cl; r14-17	1540
Antechinus bilami	2-6	10698
	r4.4-4.6 yg; n3 yr data; at pouch exit	734
	4.7; sd1.2; r3-6; n23 lt	731
	5.2; sd0.9; r3-6; n15 lt	731
	7 cl	1540
	7.5 emb	1540
Antechinus flavipes	4 yg	4989
	6.2 born; r3-12; n5 f	11912
	8.7 yg; sd1.44; r4-10; n13 lt	10108
	9.0 yg; sd1.56; r4-10; n20 lt	10108
	max 10	2841
Antechinus godmani	1-6 yg; n3 lt	11161
Antechinus leo	7 yg; n1 lt	11160
Antechinus macdonnellensis	mode 5	6344
	5.88 yg; r5-6; n8	11922
	max 6	11915a
	20.5 cl; r12-30; n10	11922
Antechinus melanurus	3.5 yg; r3-4; n4	2838
Antechinus minimus	6 yg; n1 lt	11425
	6 yg; n1 lt	4043
	6-8 yg	11746, 11748
	7.4; n8 f	11392
	8 yg ea; n12 lt	1935
Antechinus naso	3 yg, 4 swollen teats; n2 f	2838
Antechinus rosamondae	mode 6-8	6344
	6.3 yg; mode 8; r1-8; n10 f	11922a
Antechinus stuartii	3-5	11913
	5.025; r2-8; n40 lt	9804
	5.8 cl/side, 3.5 emb/side; n67 f; captive	9757
	6.1 cl/side; sd1.4; n100 sides, 53 f	9758
	6.4 cl/side, 4.8 emb/side; n22 f; wild mated	9757
	6.8; r3-8; n4 lt	6825
	6-8 yg	2554
	7.16; r0-10; n12 lt	11912
	7.17; r2-10; n107 lt	1935
	7.5	11886
	7.7 yg; se0.18; r7-8; n7 lt	610
	8 yg ea; n8 f	9757
	9-17 cl	9761

Antechinus swainsonii	7.5; mode 8; r3-8; n90 lt	1935
	8 yg; n1 lt	11395
	8.5; r8-9; n2 lt	11721
	9.33; r8, 8 yg, 12 emb; n3 lt	4043
Dasycercus cristicauda	5.2; r5-6; n3 lt	7280
	6-8	3305
	7-8	11903
Dasyuroides byrnei	3.4 wean; n10 lt	3601, 7181
	4.3 yg; r2-6; n10 lt	3601, 7181
	4.8; mode 6; r1-7; n79 lt	425
	mode 5	6697
	5.3 yg; n15 lt	423
	5.5 yg; sd1.1; n35 lt	3326
	6 yg; n1 lt	5116
	max 17 bom	11914
Dasyurus geoffroii	mode 6; r3-6; n9 lt, 6 f	9775
Dasyurus hallucatus	r4.4-5.6; n3 yr data; at pouch exit	734
	6.1 yg; sd2.2; n9 f	9618
	6.3 yg; sd2.2; n4 f	9618
	6.4 yg; sd1.1; n21 lt	732
	7.8 yg; sd0.4; n22 f	9618
	8 yg *ea*; n2 lt	3306
	17.0 blastocysts; sd5.4; n4 f	9618
Dasyurus maculatus	r1-3 yg; n6 lt	2028
	3.5 yg; r2-5; n2 lt	9802a, 9803
	4-6	10917
Dasyurus viverrinus	1-6	7221
	4.3 yg; r3-5; n3 lt	11511
	4.7 yg; mode 6; r1-6; n7 lt	3325
	5.8 yg; mode 6; r5-6; n12 lt	4040
	6	4754
	6 yg	1387
	6 yg; n1 f	3297
	20-35 oocytes	4755
Ningaui spp	5 yg; n1 f	3313
	6.2; mode 7; r2-7; n6 lt	3163
	7 yg; n1 lt	5348
Ningaui yvonneae	7 yg *ea*; n2 lt	2082a
Phascogale calura	mode 6-8	6344
Phascogale tapoatafa	6.4 yg; r3-8; n13 lt	2189
Planigale gilesi	4.5; r3-7; n4 lt	11646
	6.1; r3-10; n13 lt	8992
Planigale ingrami	3.0 wean; n3 lt	7072a
	4-6 yg	355
	4-12 yg	4587
	7 yg; n1 lt	3307
	10; r8-12; n2 lt	2309
	15 yg; n1 nest	3308
Planigale maculata	5 yg; n1 f	6826
	8.0 yg; r4-12; n21 lt	10699
	8.1 yg; r5-11; n9 lt	424
Planigale tenuirostris	6 yg; n1 f	11498
	6.0; r4-9; n10 lt	8992
Sarcophilus harrisii	2 yg; n1 lt	4462
	2 yg; n1 f	3751
	2; r1-3; n2 lt	4040
	2.88 yg; mode 4; r1-4; n53 lt	4132
	4 yg; n1 lt	3298, 3304
	mode 4 71%; r1-4; n21 lt	5077
	21 oocytes; n1 f	3353
	max 23 ovulations	5076
	39 ovulations; r11-56; n6 f	5077
Sminthopsis crassicaudata	r1-9 bom; n9 lt	11928
	5.4; se0.3; r3-8; n24 lt, 50-70 d	7552

	5.47 yg; n105 lt	3843
	6-8; n8 lt	6886
	6.09; r3-10; mode 5-6; n11 lt	3131
	7.0 wean; sd5.7; n120 lt	778
	7.3 yg or swollen teats; mode 8; r3-10; n23 museum f	3843
	7.5; se0.5; r5-10; n14 lt, 0-9 d	7552
	8.1; r7-9; n10 lt, <10 d	7552
	8.5 emb; r7-10; n2 f	7552
Sminthopsis laniger	4.5 yg; r3-6; n2 f	6436
	6 yg ea; n3 lt	11918
	7; n1 lt	10917
Sminthopsis leucopus	8 yg; n1	11923
	13 cl; r11-14; n3	11923
Sminthopsis longicaudata	3.0; r1-5; n2	11927
Sminthopsis macroura	1-8	3838
	6.0 yg; r1-8; n89 lt	11920
	14.7 emb; 8.4; sd2.8; n10 right horn, 6.3; sd4.5; n10 left horn	9761a
	20.7 cl	11920
Sminthopsis murina	6.75 yg; r1-10; n8 lt	3439
	10 yg; n1 lt	2841
Sminthopsis ooldea	7; r5-8; n7 lt	426
SEXUAL MATURITY		
Antechinus apicalis	f/m: 10-11 mo	11915
Antechinus bilarni	f/m: 1st mate: 11 mo	731
Antechinus flavipes	f/m: 1st mate: 11 mo	11913
Antechinus macdonnellensis	f/m: ~11 mo	11915a
	f/m: 1 yr	10698
Antechinus minimus	f/m: 1st mate: 11 mo	11392
Antechinus ningbing	f/m: 10-11 mo	11919
Antechinus rosamondae	f/m: ~10 mo	11922a
Antechinus stuartii	m: 9-10 mo	11912
	f/m: 1st mate: 11 mo	11913
	f/m: 320 d	6825
Antechinus swainsonii	f/m: 1st mate: 10-11 mo	11721
Dasycercus cristicauda	f/m: 1 yr	10698
Dasyuroides byrnei	f: 1st mate: 238 d	423, 425
	m: 1st mate: 216 d	425
	f/m: 8-11 mo	10698
	f/m: 10-11 mo	7181
Dasyurus geoffroii	f: 12 mo	376
Dasyurus maculatus	f/m: 1st mate: 11 mo	6346
	f/m: 1 yr	2028
Dasyurus viverrinus	f/m: 1st mate: 11 mo	6346
Phascogale tapoatafa	f/m: 1 yr	10698
Planigale maculata	f: 7.0 g	10699
	m: 8.5 g	10699
	f/m: 1st mate: 290 d	424
Sarcophilus harrisii	f: ~4-4.9 kg; n19	5077
	f/m: 1st mate: 2 yr	4134
Sminthopsis crassicaudata	f: 1st estrus: 100-120 d	1247
	f: 1st estrus: 115 d	3131
	f: 1st mate: 6 mo	778
	f: 1st birth: 18.5 wk; n1	3131
	f: 1st birth: 158.5 d; r152-165; n2	3843
	m: 1st mate: 159 d	3131
	m: 160 d	1247
Sminthopsis laniger	f: 1st estrus: 11.5 mo	11918
Sminthopsis leucopus	f/m: 11 mo; n7	11923
Sminthopsis macroura	f: 1st estrus: 119 d	3838
	f: 1st mate: bimodal 86-159 d, 12.5-16.0 g; or 185-262 d, r18.0-22.5 g	11920
	m: 1st sperm: 141-350 d; n60	11921
	f/m: 1st mate: 135 d	3838
Sminthopsis murina	m: 90-100 d	3439

Sminthopsis ooldea	f/m: 10 mo	426
Sminthopsis virginiae	f: 1st estrus: 230 d	10642
CYCLE LENGTH		
Antechinus apicalis	estrus: 4 d	11915
Antechinus macdonnellensis	estrus: 18.7 d; r10-27; n15	11922
Antechinus stuartii	estrus behavioral: 6.2 d; max 13	11912
	estrus histological: 19.33 d; sd4.42; n41 f	9757
Dasyuroides byrnei	60.3 d; sd7.1; n15	3326
Dasyurus maculatus	21 d	9803
Dasyurus viverrinus	37.4 d; se0.65; r28-43; n27	4803
	37.5 d; se0.7; r32-43; n20	4803
Planigale gilesi	estrus: 3 d	7983
	mode 21 d; r15-22; n7 cycles, 3 f	8992
Planigale tenuirostris	estrus: 1 d	7983
	mode 33 d; r31-34; n4 cycles, 4 f	8992
Sminthopsis crassicaudata	estrus: 1-3 d	1247
	estrus: 1-3 d	3131
	~28 d	1247
	mode ~30 d; r29-45	3131
	31.1 d; ±0.7; r25-37; n25 cycles; 17 f	3843
Sminthopsis laniger	30-51 d	10698
Sminthopsis leucopus	43 d; r40-46; n2 cycles, 2 f	11925
Sminthopsis longicaudata	34.4 d; r29-45; n5	11927
Sminthopsis macroura	23.25 d; r19-30; n25	11920
	26.2 d; ±0.5; r21-38; n32 cycles, 18 f	3838
Sminthopsis murina	24 d; r20-30; n5 f	3439
Sminthopsis virginiae	r29-40 d; n3 f	10642
GESTATION		
Antechinus apicalis	44-53 d	11915
Antechinus flavipes	23-27 d	11912, 11913
	~1 mo	2841
Antechinus macdonnellensis	> 38 d	11915a
	40 d; r36-44; n3	11922
	51 d; r45-55; n3; with sperm storage?	11922
Antechinus minimus	27.8 d; sd2.2; r26-30; n7; field capture to birth	11746
	30.6 d; sd1.5; r29-32; n3; lab	11746
Antechinus ningbing	est 45-52 d	11919
Antechinus rosamondae	40-50 d	10698
	49.7 d; r38-62; n10	11922a
Antechinus stuartii	diapause: 1-5 d	9755
	26-35 d; last mating to birth	11912, 11913
	27.2 d; se0.55; sd1.83; mode 26-28; r25-31; n26 f	9754, 9755, 9757
	31 d	2553
	32-34 d	11396
Antechinus swainsonii	28 d	2553
	31.5 d; r28-35; n2	11721
Dasycercus cristicauda	~30 d	7280
	~30 d; n1	3305
	35-44 d; n6	11914
Dasyuroides byrnei	r30-35 d; n32	11914
	r30-36 d; n10	423, 425
	32 d	6697
	32.5 d; sd1.8; r28-36; n35	3326
	33 d 18 h 58 min; n1	5116
	median ~35 d; r34-36; n10	7181
Dasyurus geoffroii	16-23 d	376
Dasyurus maculatus	21 d	9803
	~3 wk	3301
Dasyurus viverrinus	20 d; ±2	1387
	20.3 d; se0.63; r19-22; n4	4803
	20.5 d; se0.8; n6	4803
	21.02; sd2; r20-24; n4	3325
	est 3 wk; n3 lt	11511
Ningaui spp	14.7 d; r13.5-21.25; n5	3163

Phascogale tapoatafa	27.5 d; r26-29; n2; capture to birth	2189
	30 d	10698
Planigale gilesi	15.5 d; r14.5-16.5; n3	11646
Planigale maculata	19-20 d; n2	424
	20.5 d; r20-21; n2	11158
Planigale tenuirostris	mode 19 d; r18.5-19.5; n5 f	8992
Sarcophilus harrisii	~3 wk	5077
	31 d	4134
Sminthopsis crassicaudata	13 d	7552
	13.9 d; r13-16; n10	3843
	~16 d	3131
	16-18 d	1247
Sminthopsis laniger	11-12 d	10698
	est ≤12 d; weight changes	11918
Sminthopsis leucopus	est 20 d; no epithelial cells in urine to milk secretion; 3 wk	11923
Sminthopsis longicaudata	est <15 d; young present 17, 19 d after mating	11927
Sminthopsis macroura	10.7 d; sd0.67; most 10.5-11.0 d; r9.5-12.0	9761a
	r11-20 d; n74 lt	11920
	12-13 d; n8	3838
Sminthopsis murina	12.5 d; se0.28; r11-13.5	3439
Sminthopsis ooldea	≤ 21 d; pairing to birth	426
Sminthopsis virginiae	13-16, 16-19, 17-20 d; n3	10642
LACTATION		
Antechinus apicalis	wean: 4 mo	11915
Antechinus bellus	teat attachment: 4-5 wk	3520
Antechinus bilarni	wean: min 4 mo	731
Antechinus flavipes	teat attachment: 31 d	11912
	pouch exit: 30-36 d	10108
	nurse: 3-4 mo	10108
	wean: 4 mo	11912
Antechinus macdonnellensis	wean: 3-4 mo	11915a
	nurse: 14 wk	11922
Antechinus minimus	teat attachment: 2 mo	11746, 11748
	solid food: 8 wk	11425
	nurse: 12 wk	11425
	nurse: 80-90 d	11746
Antechinus ningbing	wean: ~16 wk	11919
Antechinus rosamondae	solid food: 70 d	11922a
	wean: 90-120 d	11922a
	wean: 4 mo	10698
Antechinus stuartii	teat attachment: 30-40 d	11912
	teat attachment: ~5 wk	11396
	teat attachment: 40 d	6825
	nurse: 90 d	6825
	nurse: min 3 mo	11396
	wean: 3 mo	11886
	wean: 3-4 mo	11912
Antechinus swainsonii	teat attachment: 8 wk	11721
	solid food: 12 wk	11721
	nurse: 14 wk	11721
Dasycercus cristicauda	teat attachment: 55-60 d	7280
	teat attachment: ~58 d; n1	3305
	teat attachment: ~1 mo	10917
	solid food: 108 d	7280
	wean: 121 d	7280
	independent: 3.5 mo	3305
Dasyuroides byrnei	teat attachment: 56 d	423, 425
	teat attachment: 56 d; r50-59	7181
	solid food: 95-96 d	7181
	wean: 100-105 d	423, 425
	nurse: 127 d; r115-144	7181
Dasyurus geoffroii	pouch exit: 62-72 d	9775
	pouch exit: 70-90 d	9378
	wean: 120+ d	9378

Dasyurus hallucatus	teat attachment: 8 wk	3306
	nurse: 6-7 mo	732
Dasyurus maculatus	teat attachment: 47-50 d	9803
	solid food: 17 wk	2028
	wean: 120-150 d; n2 lt	9803
	independent: 4.5 mo	3301
Dasyurus viverrinus	teat attachment: 9 wk; n1	3297
	teat attachment: 60-65 d	1387
	solid food: 107-112 d	7221
	wean: 102-142 d singletons; -200 d larger lt	7221
	nurse: 4 mo	9470
	independent: 4.5 mo	3297
	wean: 5.5-6 mo	1387
Ningaui spp	teat attachment: 42-44 d	3163
	solid food: 70 d	3163
	wean: 76-81 d	3163
Phascogale tapoatafa	teat attachment: 7-8 wk	2189
	wean: ~16 wk	2189
Planigale gilesi	teat attachment: 37 d	11646
	solid food: 65 d	11646
	solid food: 66 d; n1 lt	6519
Planigale maculata	teat attachment: 26 d; r24-28; n2	11158
	teat attachment: 28 d	424
	solid food: 55 d	424
	solid food: 57 d	11158
	independent: 70 d	424
	independent: 72 d	11158
Planigale tenuirostris	teat attachment: 36 d	8992
	nurse: 95 d	8992
Sarcophilus harrisii	teat attachment: 15 wk	3304
	pouch exit: 105 d	4132, 4134
	solid food: 5 mo	3304
	wean: ~8 mo	4132, 4134
Sminthopsis crassicaudata	teat attachment: 40 d	3843
	teat attachment: 40-43 d; n4 lt	3131
	solid food: 60 d	3843
	wean: 65-69 d; n3 lt	3131
	wean: ~70 d	1247
	wean: ~70 d	3843
Sminthopsis laniger	wean: 3 mo	11918
Sminthopsis leucopus	wean: 2.5 mo	11923
Sminthopsis macroura	teat attachment: 40 d	3838
	solid food: 60 d	3838
	wean: 70 d	3838
	wean: 70 d	11920
Sminthopsis murina	teat attachment: 34 d; n1 lt	3439
	solid food: 63 d; n1 lt	3439
Sminthopsis ooldea	wean: ~70 d	426
Sminthopsis virginiae	teat attachment: 55-56 d	10642
	solid food: 74-78 d	10642
	wean: 80-90 d	10642
INTERLITTER INTERVAL		
Dasyuroides byrnei	161-173 d	425
Planigale maculata	40 d; n1 with loss of lt	424
Planigale tenuirostris	min 91 d	8992
Sminthopsis crassicaudata	mode 85 d; r82-210	778
	~12 wk	3131
	min 12 wk	3843
	~97 d; min 12 wk	3131
	4 mo	6886

LITTERS/YEAR		
Antechinus apicalis	1	11915
Antechinus bilarni	1	731
Antechinus flavipes	1	10108
Antechinus minimus	1	11748
Antechinus stuartii	1	11912
	1	11886
Dasycercus cristicauda	1/season	10698
Dasyuroides bymei	max 2 lt/season	425
Dasyurus hallucatus	1	732
Ningaui spp	1-2	3163
Phascogale tapoatafa	1	2189
Planigale maculata	several	424
Sarcophilus harrisii	1	5077
Sminthopsis crassicaudata	1-3	10129
	2	7551
	est 2	7552
Sminthopsis laniger	>1/season	11918
Sminthopsis murina	max 2	3439
Sminthopsis macroura	2	11920
SEASONALITY		
Antechinus apicalis	sperm in urine: Jan-mid Apr; captive	11915
	mate: Mar-Apr; captive	11915
	birth: Apr; sw Australia	11915
Antechinus bellus	fertile m: mid June-Aug; NT, Australia	1540
	fertile m: mid June-mid Aug; NT, Australia	10703
	mate: mid-end Aug; NT, Australia	3520
	birth: end Sept-early Oct; NT, Australia	1540
	birth: mid-end Sept; NT, Australia	3520
Antechinus bilami	pouch/scrotal development: June-Aug; NT, Australia	731
	mate: May-mid July; NT, Australia	1540
	mate: June-July; NT, Australia	731
	birth: Aug-Sept; NT, Australia	731
	lact f: Aug-Dec; NT, Australia	731
Antechinus flavipes	m death: Sept; QLD, Australia	10108
	m death: end Sept; QLD, Australia	2841
	mate: June-July; lab	11913
	mate: end July; ACT; mid Aug; NWS, Australia	2552a
	mate: end Aug-early Sept; QLD, Australia	10108
	mate: end Sept; QLD, Australia	2841
	birth: July-Aug; lab	11913
	birth: end Sept-early Oct; QLD, Australia	10108
	birth: end Oct; QLD, Australia	2841
	lact f: Oct-Jan, QLD, Australia	10108
Antechinus godmani	birth: July-Aug; ne QLD, Australia	11161
Antechinus leo	birth: early Nov; QLD, Australia	11160
Antechinus macdonnellensis	birth: late July-early Sept; central Australia	11922
	birth: Aug-Sept; Australia	11915a
Antechinus melanurus	mate: yr rd; few data; Mt Enimbari, New Guinea	2838
Antechinus minimus	prostate, cowpers: peak June-July; VIC, Australia	11747
	spermatogenesis: Feb-May; VIC, Australia	11747
	testes: peak May; VIC, Australia	11747
	m death: end July-early Aug; VIC, Australia	11747
	mate: May; VIC, Australia	11392
	preg f: Dec; n1; TAS, Australia	4043
	birth: July; VIC, Australia	11392
	birth: July-Aug; VIC, Australia	11746, 11748
	lact f: Aug; n1; captive	11425
Antechinus naso	mate: yr rd; few data; Mt Enimbari, New Guinea	2838
Antechinus ningbing	mate: June; WA, Australia	11919
Antechinus rosamondae	mate: Sept; nw WA, Australia	11922a
	birth: Oct-Nov; nw Australia	6346
	birth: Nov; nw WA, Australia	11922a

Antechinus stuartii	spermatogenesis: fails May; VIC, Australia	5694
	testes: large June-Aug; ACT, Australia	11913
	mate: July-Aug or Aug-Sept; 2 stocks, lab	9757
	mate: July-Sept; lab; ACT, Australia	11912, 11913
	mate: mid July-Sept; ACT, Australia	2553
	mate: early Aug; NSW, Australia	6825
	mate: early-mid Aug; NSW, Australia	2552a
	mate: mid-end Sept; QLD, Australia	11886
	birth: Aug-Sept; NSW, Australia	4989
	birth: Aug-Oct; ACT, Australia	11913
	birth: end Aug-early Sept or end Sept-early Oct; 2 stocks, lab	9757
	birth: end Sept; se Australia	6384
	birth: 3rd wk Sept-early Oct; e Australia	1194
	wean: Feb; QLD, Australia	11886
Antechinus swainsonii	mate: May-Sept; mainland, Australia	11721
	mate: May-Sept; ACT, Australia	2553
	mate: mid July; VIC, Australia	7047
	mate: Sept-Oct; TAS, Australia	11721
	birth: mid Aug; VIC, Australia	11395
	birth: end Aug; se Australia	6384
	lact f: ?-Feb; w TAS, Australia	4043
	wean: Dec; se Australia	6384
Dasycercus cristicauda	mate: end June-July; captive	3305
	birth: June-July; central Australia	6346
	repro: June-Sept; central Australia	11903
Dasyuroides byrnei	mate: Apr-Dec, 75% May-July; lab	423, 425
	mate: Dec-Apr; captive	7181
	birth: May-Oct; captive	6697
	birth: June; e central Australia	11495
Dasyurus geoffroii	mate: Apr; captive	353
	birth: May-Sept, peak June-July; n58; WA, Australia	10196
Dasyurus hallucatus	lact f: July-Sept; NT, Australia	732
	lact f: July-Jan; WA, Australia	9618
Dasyurus maculatus	mate: June-July; VIC, Australia	3301
	birth: ~May; Australia	10917
	birth: July-Aug; VIC, Australia	3301
Dasyurus viverrinus	testes: large Mar-June, small Nov-Feb; TAS, Australia	1388
	f cycles: May-Aug; se Australia	4803
	mate: peak end May-early June; TAS, Australia	3325
	mate: May-July; Australia	9470
	mate: May-June; TAS, Australia	3846
	lact f: July 5-26; ne TAS, Australia	4040
Ningaui spp	birth: end Oct-early Dec; n5; captive	3163
Ningaui ridei	repro: Sept-Feb; NT, Australia	7983
Ningaui timealeyi	repro: Sept-Mar, peak Nov-Jan; Australia	7983
Phascogale calura	mate: last 3 wk July; WA, Australia	1190
Phascogale tapoatafa	birth: end July-early Aug; VIC, Australia	2189
	lact: Aug-Nov; VIC, Australia	2189
Planigale gilesi	mate: end Nov-Dec; captive	11646
	birth: end Aug-mid Jan; NSW, Australia	8992
Planigale ingrami	birth: Dec-Mar; NT, QLD, Australia	4587
	lact f: Nov-Dec, Feb-Mar; n4; QLD, Australia	3307
	repro: Feb-Apr; NT, Australia	355
	repro: summer; WA, Australia	11916
Planigale maculata	mate: est yr rd; NT, Australia	424
	birth: est yr rd; NSW, QLD, Australia	11158
	pouch young: yr rd; NT, Australia	10699
Planigale tenuirostris	birth: Sept-Jan, NSW, Australia	8992
	lact f: Oct; n1; SA; Australia	11498
Sarcophilus harrisii	spermatogenesis: Apr-Aug; TAS, Australia	5077
	mate: Mar; TAS, Australia	4132
	mate: May-June; TAS, Australia	3304
	preg f: Mar-May; TAS, Australia	5076
	birth: Mar-May; TAS, Australia	5076

	birth: Mar-Apr; TAS, Australia	5077
	birth: Apr-Sept; TAS, Australia	6346
	birth: end May-early June; TAS, Australia	3304
	lact f: July; n2; ne TAS, Australia	4040
Sminthopsis crassicaudata	mate: June-July; SA, Australia	11901
	birth: May-Jan; captive	3131
	birth: est June-Dec; arid, se Australia	7552
	birth: July-Aug; VIC, Australia	3296
	birth: end July-Feb; mesic, se Australia	7552
	birth: yr rd, peak July-Jan; captive	3843, 10129
	birth: est yr rd; captive	6886
Sminthopsis laniger	birth: Aug-Dec; sw QLD, Australia	11918
Sminthopsis leucopus	sperm in urine: June-Oct; VIC, Australia	11923
	mate: Aug; VIC, Australia	11923
	birth: Aug-Sept; VIC Australia	11923
	wean: end Nov; VIC, Australia	11923
	lact f: end Aug-Oct; se Australia	8994
Sminthopsis macroura	birth: June-Feb, peak Aug-Dec; captive	11920
	birth: July-Feb; captive	3838
Sminthopsis murina	birth: Sept-Jan; NSW, Australia	3439
Sminthopsis ooldea	birth: Sept-Jan; SA, Australia	426
Sminthopsis virginiae	lact f: July-Nov; NT, Australia	7553

Order Marsupialia, Family Myrmecobiidae, Genus *Myrmecobius*

The 300-700 g, diurnal, solitary, and endangered numbat, *Myrmecobius fasciatus*, feeds on ants and termites in Western Australia (10809). Females have 4 teats and lack a well-developed pouch (1539, 3521, 9852, 10698). Details of milk composition (4066) are available.

LITTER SIZE		
Myrmecobius fasciatus	3.4 yg; mode 4; r2-4; n8 lt	1538, 1539
	4 yg; n1 f	1827
	4 yg; n1 f	3521
LACTATION		
Myrmecobius fasciatus	independent foraging: 6 mo	1538, 1539
	wean: 7 mo	10698
	wean: 9 mo	4066
SEASONALITY		
Myrmecobius fasciatus	birth: Jan-May; sw WA, Australia	1538, 1539
	birth: end Jan; sw WA, Australia	4066
	lact f: Mar-Oct; few data; WA, Australia	3521
	lact f: May; n1; WA, Australia	1827
	repro: June-July; WA, Australia	11901

Order Marsupialia, Family Thylacinidae, Genus *Thylacinus*

The Tasmanian wolf, *Thylacinus cynocephalus*, is either extinct or restricted to a small part of Tasmania, Australia (10809). This nocturnal, possibly solitary, carnivore ranges from 15 to 25 kg. Females have 4 teats (8617). A pouch is present in both sexes (8617). Anatomical details are available (8298).

LITTER SIZE		
Thylacinus cynocephalus	mode 2-3; max 4	10698
	3-4	4130
SEASONALITY		
Thylacinus cynocephalus	small juveniles: yr rd, peak May-Aug; TAS, Australia	4130

Order Marsupialia, Family Notoryctidae, Genus *Notoryctes*

The solitary and fossorial *Notoryctes typhlops* is a 40-70 g, Australian insectivore. The pouch, present in both sexes, opens posteriorly (10423). Females have 2 mammae. Nothing has been recorded about its reproduction, but its anatomy has been described (10424, 10425, 10601).

There are no tabular data for *Notoryctes*.

Order Marsupialia, Family Peramelidae

Bandicoots (*Chaeropus*, *Echymipera*, *Isoodon*, *Microperoryctes*, *Perameles*, *Peroryctes*, *Rhynchomeles*) are 200-2300 g, terrestrial, nocturnal omnivores from Australia, Indonesia, and New Guinea. *Chaeropus* may be extinct. The New Guinea and Indonesian genera *Echymipera*, *Microperoryctes*, *Peroryctes*, and *Rhynchomeles* are little known. Males are generally larger than females (286, 6670). In addition to the usual choriovitelline placenta, bandicoots have a chorioallantoic placenta (3354, 3356, 4750, 4751, 4753, 4763, 5075, 8195, 8294, 10710, 10711).

The pouch opens posteriorly (6663, 10917). Neonates have deciduous claws and open their eyes at 44-49 d (3685, 4584, 6663, 6665, 6666, 6699, 6825). Females have 6-10 teats (4233, 8798, 10698, 10917).

Details of anatomy (4752, 4753, 4923, 5073, 6671, 8000, 8291, 10858), endocrinology (1618, 1619, 2214, 3669, 3670, 3672, 3676, 3679, 3684, 3687, 3688, 3689, 3690, 5073, 5351), sex-chromosome mosaicism (1906a, 1908, 3492, 4519, 4520, 11440), primordial germ cells (11028), spermatogenesis (6931, 9793a), ovulation or conception (6673), and milk composition (7220) are available.

NEONATAL MASS		
Isoodon macrourus	188.3 mg; r148-220; n6	6699
Isoodon obesulus	350 mg	4584
Perameles gunnii	250 mg	4584
Perameles nasuta	237 mg; r209-254; n15	6663, 6665
NEONATAL SIZE		
Isoodon macrourus	CR: 13.83 mm; r13.5-14.5; n3	6699
Isoodon obesulus	11.0 mm	4584
Perameles nasuta	CR: 12.8 mm; sd0.75; n13	6663, 6665
WEANING MASS		
Isoodon macrourus	138-212.5 g	6699
LITTER SIZE		
Chaeropus ecaudatus	1 yg; n1	6662
	2 yg	10698
Isoodon macrourus	2.16 wean; ±0.5; n87 lt	3680
	2.2 yg; r1-3; n9 lt	2841
	2.5 yg; sd1.03; n16 lt, 12 f	5646
	2.6; se0.2; r1-5; n33 1st lt	3674
	2.9 yg; r1-4; n12 lt	10443
	2.9; r1-5; n44 lt	4233
	3.0 yg; ±0.15; mode 3; r1-6; n136 lt, 0-10 d	3680
	3.03 yg; 2.31 wean; n32 lt	3691
	3.07 yg; se0.11; r1-6; n112	6670, 6671
	3.07 yg; se0.12; n222 lt, 0-10 d	3681
	3.19 yg; 2.56 wean; r1-5; n27 lt	3686
	3.38 yg; r1-7; n21 f	3946
	3.4 yg; se0.2; n36, 1-5 d	3671
	3.6 yg; se0.3; n14, 1-10 d	3671
	4.0; r1-7; n8 lt	6699
	4.0 yg; se0.4; n14 births	3671
	5.05 cl; se0.23; r3-7; 4.05 emb; se0.29; r1-6; n20	6671
	5.13 cl; r3-7; n39	6671
Isoodon obesulus	1.7 yg; r1-3; n6 f	11492
	2.3 yg; r1-5; n9 lt	6663
	2.8 yg; mode 4; r1-4; n15 lt	4584
	3.07 yg; mode 3-4; r1-6; n56 lt	10448

Perameles bougainville	mode 1-2; r1-3	10809
	2 yg; n1	9194
	2-4	10917
Perameles eremiana	mode 2	10917
	2	11901
Perameles gunnii	2.1 yg; r1-5; n25 lt, 9 f	9717
	2.23 yg; r1-4; n22 lt	6663
	2.33 yg; mode 2; r1-4; n54 lt, 1-20 d	4584
Perameles nasuta	1.83 yg; n6 lt	6665
	2-3 yg	6675
	2.44 yg; r1-5; n52 lt	6663
	2.6; r1-5	6670
	2.65 yg; se0.07; r1-5; n196	6671
	3.31 cl; r2-6; n45	6671
	3.33 cl; se0.22; r2-6; 2.63 emb; se0.26; r1-5; n24	6671
SEXUAL MATURITY		
Isoodon macrourus	m: high testosterone: 179.8 d; se6.8; r127-243; n17	3677
	m: 1st sperm: 199.5 d; se8.2; r201-224; n10	3677
	m: 1st sire: 349.7 d; se14.0; r293-383; n7	3677
	f: 1st birth: 249.6 d; se11.4; r161-399; n33	3674
	f: 1st repro: ~4 mo	4584
Perameles gunnii	f: 1st mate: ~2.5-3.5 mo	4584
	m: epididymal sperm: 4-6 mo	4584
Perameles nasuta	f: 1st birth: ~4 mo	6663
	m: epididymal sperm: 150 d	6663
CYCLE LENGTH		
Isoodon macrourus	rpy to ovulation: 6 d	3678
	rpy to estrus: ~20 d	3670
	20 d; mode 18; r9-34; n39 cycles, 15 f	6668
	22.1 d; se1.6; r14-30; n10 f	3678
Perameles nasuta	rpy to estrus: 17-26 d	1907
	~20 d; n3 cycles	1907
	21 d; r10-34; n9 cycles, 8 f	6668
	26 d; r17-34; n8 cycles	5073
GESTATION		
Isoodon macrourus	12 d, 8-11 h; n1	6666
Perameles nasuta	11-13 d; n1	5073
	12.5 d; n3	10443
LACTATION		
Isoodon macrourus	teat attachment: 42.4 d; se0.5; n10 lt	3685
	pouch exit: 48.7 d; se0.7; n8 lt	3685
	pouch exit: 50 d; ±7; n6 lt	3946
	pouch exit: 8 wk	6699
	wean: 59.8 d; se0.9; n20 lt	3685
Isoodon obesulus	wean: 58 d	10448
Perameles gunnii	wean: 59-61 d	4584
Perameles nasuta	pouch exit: 48-63 d	6663
	pouch exit: 50-54 d	10443
	wean: ~67 d	5073
	wean: ~75 d	6665
INTERLITTER INTERVAL		
Isoodon macrourus	56 d; n6 lt, 1 f	3946
	61 d	3685
Isoodon obesulus	89.5 d; r51-108; n3 f	5646
Perameles gunnii	~65 d	9717
LITTERS/YEAR		
Isoodon obesulus	1	11492
	2.61/season	10448
	max 3	5646
Perameles gunnii	3.8; r3-4	4584

SEASONALITY		
Chaeropus ecaudatus	mate: May-June; Australia	10917
Isoodon macrourus	prostate: large July-Sept, small Mar-May; captive	10858
	birth: end July-early Sept; n6; captive	3676
	birth: Sept-Apr; WA, Australia	5646
	birth: yr rd, low Apr-June; QLD, Australia	3671
	birth: yr rd; QLD, Australia	4233
	birth: yr rd; captive	6699
	birth: yr rd, peak Aug-Oct, low Mar-June; captive	3681
	lact f: Sept-Feb; QLD, Australia	2841
	lact f: peak end Sept-Nov; TAS, VIC, NSW, QLD, Australia	597
	lact f: yr rd; QLD, Australia	3946
Isoodon obesulus	mate: July-Feb; TAS, Australia	4584
	birth: Sept-Oct; SA, Australia	11492
	lact f: Aug-Jan; VIC, Australia	10448
Perameles bougainville	mate: May-Aug; Bemier I, w SA, Australia	10809
Perameles gunnii	epididymal sperm: yr rd; TAS, Australia	4584
	mate: end May-Dec; TAS, Australia	4584
	birth: peak July-Nov; VIC, Australia	9717
Perameles nasuta	mate: peak spring-summer; captive	6668, 6675
	mate: yr rd; captive	1907
	lact f: May; n1; QLD, Australia	2841
	lact f: yr rd; n~50; NSW, Australia	6663

Order Marsupialia, Family Thylacomyidae, Genus *Macrotis*

The nocturnal bilbies, *Macrotis lagotis* (800-2500 g) and *M. leucura* (310-435 g), from arid Australian grasslands are fossorial when inactive (5337). These colonial omnivores feed preferentially on ants and termites in parts of their range (10174, 11491). Males are larger than females (5337) and have an extra chromosome (4013). Eyes open 43-64 d (5088). Females have 6-8 teats (10474, 10917). *Macrotis leucura* may be extinct. Details of ovarian anatomy (7037) are available.

LITTER SIZE		
Macrotis lagotis	1 yg; n1 lt	6662
	1.6 yg; r1-2; n5 lt	5088, 5089
	2 yg; n1 lt	5337
CYCLE LENGTH		
Macrotis lagotis	20.6 d; se7.3; r12-37; n14 cycles, 6 f	7037
GESTATION		
Macrotis lagotis	14.0 d; se1.4; r13-16; n4; comified cells to birth	7037
	21 d; n1	5337
LACTATION		
Macrotis lagotis	pouch exit: 67-68 d	5088
	pouch exit: 80 d; se2; n6	7037
	nurse: r21-88 d after pouch exit; n4	7037
SEASONALITY		
Macrotis lagotis	birth: Mar-May; Australia	11901

Order Marsupialia, Family Phalangeridae

Phalangers (*Phalanger*, *Trichosurus*, *Wyulda*) are 1-5 kg, arboreal herbivores in forests of Australia and New Guinea. Males are occasionally reported larger than females (3785, 5097, 8746, 10081) and may have an extra chromosome (7659). Females have larger adrenal glands than males with a distinctive zone (1553, 5696a, 9141, 11573). Estrus does not occur after birth (3784, 5610, 5613, 10127, 10130). The eyes of *Trichosurus* neonates open at 100-120 d (5014, 6676, 10998). Female *Trichosurus* have 2 mammae within a pouch (4973, 10917, 10998, 11902).

Details of anatomy (378, 3787, 5613, 8292, 10735, 11098, 11100), endocrinology (195, 1060a, 1065, 1556, 1618, 1619, 1766, 2030, 2174, 2179, 2180, 2182, 3675, 3688, 4807, 5097, 5405, 6824a, 9790, 9791, 9793, 9854, 9929, 9930, 10395a, 10397, 10398, 10805, 10990, 10992, 11308, 11571, 11572), intersexes (1059, 1060, 1061, 1062, 1063, 1064, 1067, 9863), spermatogenesis (2167, 4369a, 9208, 9746, 9793a, 10735a), prostate function (2030), semen sugars (9209), superfetation (5610), egg membranes/placentation (5082, 9853), implantation (5075), mammary glands (9854), milk composition (1058, 2141, 4072, 4100, 4101), and day length (3682) are available.

NEONATAL MASS		
Trichosurus vulpecula	180 mg; r170-190; n2	5612
	205 mg; r200-210; n2	6674
	est 230 mg from growth regression	5612
	500 mg; 2 d	8746
NEONATAL SIZE		
Trichosurus vulpecula	CR: 13 mm; n1	6674
WEANING MASS		
Trichosurus arnhemensis	470 g	10081
Trichosurus caninus	m: 666.67 g; se92.80; n3	5014
	f: 881.00 g; se42.56; n5	5014
Trichosurus vulpecula	321 g; r229-439; 170 d	750
	~600 g; 170 d	6676
	f: 1300 g	10081
LITTER SIZE		
Phalanger carmelitae	1 yg	7210
Phalanger gymnotis	1 yg	7210
Phalanger maculatus	1 yg; n1	7207
	2-4	10917
Trichosurus arnhemensis	1 10081	
Trichosurus caninus	1.08; r1-2; n13 lt	10130
Trichosurus vulpecula	1 ea; n1067 lt	750
	1.03; r1-2; n36 f	10985
	1.09 cl; r1-2; n99 lt	10127
	1.13; mode 1; r1-3; n60 lt	8798, 12128
Wyulda squamicaudata	1 ea; n4 lt	5097
SEXUAL MATURITY		
Phalanger maculatus	m: testes descend: est 8 mo; n1	7207
Trichosurus caninus	f: 1st ovulate: end 2nd yr	10130
	f: 1st birth: end 3rd yr	10130
Trichosurus vulpecula	f: large follicles: 8-10 mo	8746
	f: 1st estrus: 9 mo	8545
	f: 1st estrus: 12 mo	5613
	f: 1st birth: mode 2 yr	750
	f: 1st birth: 2 yr	2120
	f: 1st wean: 2.6 yr	5610
	f: end 1st yr	10127
	f: early 2nd yr; a few end 1st yr	3784
	m: testes large: 1925 g; n1	2810
	m: sperm: 2268 g	10985
	m: 24 mo	10998
	m: ~6 mo later than f	3784
	f/m: 15 mo	4973
	f/m: 1st mate: yr following birth	11900
Wyulda squamicaudata	f: ~2 yr; ~1.1 kg	5097
CYCLE LENGTH		
Phalanger maculatus	est 28 d	7210
Trichosurus caninus	rpy to estrus: 10-11 d	10130
	26.4 d; ±1.0; mode 23-24 47%; r23-38; n17 cycles	10130
Trichosurus vulpecula	rpy to estrus: 8.02 d; ±0.18	8545
	luteal phase: 8 d	9930
	22-23 d	5610
	mode 22-32 d; r22-58; n25 f	8545
	24.5 d; r21-30; n13 f	6674
	25.69 d; se0.31	9859

	26 d	10805
	26 d; r24-29; n4 cycles, 4 f	1858
	26.3 d; se1.46; abstract; 27.2 d; se0.84; text; n39 cycles	2181
	26.59 d; ±0.31; n49 cycles, 23 f	8545
GESTATION		
Trichosurus caninus	16.2 d; ±0.2; mode 16 60%; r15-17; n10 lt	10130
Trichosurus vulpecula	16-18 d; n10 f	8545
	17.1; se0.14; n21 lt	2181
	17.5 d; r17-18; n2	5610
	~17.5 d; r17-21; n6	6674
	≤21 d	6676
LACTATION		
Phalanger maculatus	pouch exit: est 6-7 mo	7207
Trichosurus arnhemensis	wean: 165-195 d	10998
	wean: 6 mo	10081
Trichosurus caninus	teat attachment: 81-112 d	5014
	pouch exit: most 175-200 d	5014
	wean: 240-270 d	5014
Trichosurus vulpecula	teat attachment: 94 d	2809
	pouch exit: 112-144 d	6676
	pouch exit: 121 d	2809
	pouch exit: 140 d	10127
	pouch exit: 170 d	750
	pouch exit: ~5 mo	2810
	wean: 5-6 mo	4973
	wean: 175 d	5016
	wean: 220-290 d	10127
	nurse: 6-7 mo	2810
Wyulda squamicaudata	wean: >8 mo	5097
INTERLITTER INTERVAL		
Trichosurus vulpecula	9.0 mo; se0.46; abstract; 9.4 mo; se0.36; text; n41	2181
LITTERS/YEAR		
Trichosurus arnhemensis	1.8	10081
Trichosurus caninus	1	5014
Trichosurus vulpecula	1	750
	1-2	6674
	1-2	10985
	1-2	5610, 5611
	2	2810
Wyulda squamicaudata	1	5097
SEASONALITY		
Phalanger carmelitae	lact f: Feb-Apr, Aug-Sept; New Guinea	7210
Phalanger gymnotis	repro: yr rd; New Guinea	7210
Phalanger orientalis	repro: June-Oct; New Guinea	7210
Trichosurus arnhemensis	mate: yr rd; ne Australia	5690
Trichosurus caninus	birth: peak Mar-May; NSW, Australia	5014
Trichosurus vulpecula	sperm: few summer, some aspermy each mo; New Zealand	5610
	sperm: yr rd; NSW, Australia	10127
	testes large, sperm: yr rd; New Zealand	3784, 3787
	estrus: Apr-May; New Zealand	5613
	mate: Mar; SA, Australia	6674
	mate: June; NSW, Australia	6674
	mate: June; n1; SA, Australia	11900
	mate: peak Sept-Oct, minor peak Apr; New Zealand	3784
	mate: yr rd; n Australia	2214
	mate/birth: May; New Zealand	1298
	preg f: Mar-Apr, Aug-Sept; NSW, Australia	1058
	birth: Mar-Apr, Sept; ACT, Australia	2809
	birth: peak Mar-Apr, none Jan-Feb; SA, Australia	9852
	birth: Mar-May, Aug-Oct; VIC, Australia	8746
	birth: Mar-June, peak Apr; NSW, Australia	10127
	birth: end Mar-mid Apr; n4; captive	3675
	birth: peak Apr-May, some mid Aug-Nov; ACT, Australia	2810
	birth: Apr-June 99%; New Zealand	750

	birth: Apr-June, peak July-Sept; New Zealand	2120
	birth: yr rd, 86% Mar-May; New Zealand	5612
	birth: yr rd, peak Mar-May; zoo	4973
	birth: yr rd, peak Nov-Mar; SA, Australia	9469
	birth: yr rd; New Zealand	10985
	birth: yr rd; n60; zoo	12128
	pouch exit: ~end Sept; SA, Australia	11901
Wyulda squamicaudata	birth: mid Feb-mid Aug; n9; WA, Australia	5097

Order Marsupialia, Family Burramyidae

Pygmy possums (*Acrobates*, *Burramys*, *Cercartetus*, *Distoechurus*) are 6-100 g, nocturnal, arboreal omnivores from heath, forests, and shrublands of New Guinea and Australia. Sexual dimorphism is variable (3314, 10081). Embryonic diapause and postpartum estrus occur (1164, 1856, 10483, 11453, 11455, 11456). Neonates open their eyes at 37-48 d (*Burramys* 2582, 5691, 10998), 41 d (*Cercartetus* 11453), or 60-76 d (*Acrobates* 3314, 10998, 11454). Pygmy possums usually have 4-6 teats in a forward opening pouch (2582, 2837, 3314, 10123, 10698, 10917, 10998, 11393, 11394, 11901), although *Distoechurus* has been reported to have 1 or 2 (10684, 11455). Details of anatomy (1856, 11455, 11456, 11457) are available.

NEONATAL MASS		
Acrobates pygmaeus	17.8 mg	11454
Cercartetus concinnus	12 mg	10998
Cercartetus nanus	19.8 mg	11453
NEONATAL SIZE		
Acrobates pygmaeus	CR: 5.5 mm	11454
Burramys parvus	CR: 7.2-7.5 mm	6785
Cercartetus nanus	CR: 5-7 mm	11453
WEANING MASS		
Acrobates pygmaeus	1.7 g; pouch exit, ~65 d	11454
	7 g	3314
	8-8.5 g; 100 d	11454
Burramys parvus	~22 g; 7-8 wk	5691
Cercartetus caudatus	11-12 g	435
	11-13 g	2838
Cercartetus nanus	2.2-2.8 g; pouch exit, 33-37 d	11453
	8-12 g	10474
	12-13 g; 60-65 d	11453
LITTER SIZE		
Acrobates pygmaeus	2.52 wean; mode 4; n33	11454
	2.7 yg; r1-4; n32	3314
	2.81 yg; mode 2-3; r1-4; n59	11454
	3.0 enlarged teats; r2-4; n28	3314
	3.27 enlarged teats; mode 4; r1-4; n51	11454
	3.5 emb; r2-5; n15 f	11455, 11456
Burramys parvus	2.25; r1-4; n4 lt	5691
	3.6 yg; sd0.9; n441	6785
	4 yg ea; n2 lt	2582
Cercartetus caudatus	2.0 yg; mode 2; r1-3; n8 lt	2838
	2.7 yg; r1-4; n7	435
Cercartetus concinnus	3.33 wean; r1-6; n7 nests	11455
	4 yg; r3-6; n4 f	1856
	4.25 emb; r4-5; n4 f	1856
	4.62 yg; mode 5; r3-6; n21 lt; lt of 10 excluded	11455
	5 yg; r1-6	10917
	5.29 enlarged teats; mode 6; r4-6; n7 lt	11455
	6 yg; n1	1164
	6 yg; n1 f	1650
	8 emb; r6-12; n7 f	11455
Cercartetus lepidus	3.25 yg; r2-4; n4 lt	11455, 11455a
	3.33 yg; r2-4; n3 lt	4043a
	4 yg; n1 f	2082a
Cercartetus nanus	3.75 yg; mode 4; r2-5; n53 lt	11453

	3.90 wean; mode 4; r3-4; n10 nests	11453
	3.92 enlarged teats; mode 4; r3-6; n55 lt	11453
	5 yg; n1	11394
Distoechurus pennatus	1 yg	10684
	1 yg; n1	11924
	1 yg; n1 f	11455
	2 emb; n1 f	11455
SEXUAL MATURITY		
Acrobates pygmaeus	f: 1st mate: 8 mo	3314
	m: testes large: 7-12 mo	3314
	m: est 9-18 mo; n2	11457
Burramys parvus	f: pouch development: 2nd yr	5691
	f: 1st mate: ~10-11 mo	6785
	m: large scrota: 8 mo	5691
	m: 1st mate: ~10-11 mo	6785
Cercartetus caudatus	f: 1st birth: 460-470 d; n2	435
Cercartetus nanus	f: 1st preg: 4.5-9 mo	11453
	m: large scrota: 4.5-9 mo	11453
CYCLE LENGTH		
Burramys parvus	20.3 d; r17-26.5	6785
GESTATION		
Burramys parvus	14-16 d; n1	5691
Cercartetus concinnus	min 51 d	1164
LACTATION		
Acrobates pygmaeus	pouch exit: 50-60 d	3314
	pouch exit: ~65 d	11454
	wean: 95-100 d	3314
	wean: 100 d	11454
Burramys parvus	pouch exit: 3-4 wk	5691
	pouch exit: min 24 d	2582
	pouch life: est ~35 d	6785
	wean: 55-60 d (table), 8-9 wk (abstract)	5691
	1st trapped: ~71 d	6785
Cercartetus caudatus	1st pouch exit: ~34 d; ±5	435
	independent: 92 d; ±10	435
Cercartetus concinnus	wean: ~50 d	1164
Cercartetus lepidus	independent: 3 mo	10474
Cercartetus nanus	pouch exit: 33-37 d	11453
	wean: 60-65 d	11453
INTERLITTER INTERVAL		
Acrobates pygmaeus	r68-161 d; n11	11454
Cercartetus nanus	r56-135 d; n20	11453
LITTERS/YEAR		
Acrobates pygmaeus	2	3314
	2	11456
Burramys parvus	1	2582
	1	6785
Cercartetus caudatus	est 2	435
Cercartetus concinnus	>1	1856
SEASONALITY		
Acrobates pygmaeus	birth: July-Jan, peak Aug, Nov; VIC, Australia	3314
	birth: mid July-Feb; VIC, Australia	11454, 11456
Burramys parvus	birth: Jan, Apr, Sept, Nov; n4; captive	5691
	birth: end Oct-mid Nov; VIC, Australia	6785
	birth: Nov-Dec; n4; NSW, Australia	2582
Cercartetus caudatus	lact f: May-Dec, perhaps yr rd, low mid May-July; New Guinea	2837
Cercartetus concinnus	preg/lact f: est yr rd; WA, Australia	1856
	lact f: Jan, July, Sept; SA, Australia	10917, 11901
	lact f: May-Feb; SA, VIC, Australia	11455
	lact f: July-May; WA, Australia	11455
Cercartetus lepidus	birth: Sept-Jan; TAS, Australia	10474
	birth: yr rd; VIC, Australia	11455a
Cercartetus nanus	birth: Apr-Aug; Australia	10081
	birth: yr rd, peak Oct/Nov-Mar/Apr; VIC, Australia	11453

Order Marsupialia, Family Petauridae

Petaurids (*Dactylopsila* 250-400 g; *Gymnobelideus* 100-165 g; *Petaurus* 100-710 g; *Pseudocheirus* 440-2000 g; *Schoinobates* 900-1800 g) are nocturnal, arboreal, gliding omnivores from Australian and New Guinean forests. *Gymnobelideus* is endangered (10080). Males may be larger than females in *Petaurus* and *Pseudocheirus*, but not in *Gymnobelideus* (10080, 10081). Males have a large prostate, 2 pairs of bulbourethrals, and 3 pairs of paracloacal glands (10124, 10735). A copulatory plug is present after mating in *Petaurus* (10120). Females have 2-4 teats (3353, 4192, 10120, 10698, 10917, 10998). *Schoinobates* females retain the entire Wolffian duct (10141). Neonates open their eyes 74-82 d (*Petaurus* 10120), 91-106 d (*Pseudocheirus* 4192, 10802), or 121 d (*Schoinobates* 10998). Details of anatomy (363, 5083, 8293, 10124), endocrinology (1618, 1619, 9790), spermatogenesis (534), placentation (549, 9853), and milk composition (7880) are available.

NEONATAL MASS		
Petaurus breviceps	194 mg; n2	1856, 10120
Pseudocheirus herbertensis	180 mg; n1	4192
Pseudocheirus peregrinus	300 mg; n1	10802
Schoinobates volans	273 mg; n1 preserved	549
NEONATAL SIZE		
Pseudocheirus peregrinus	TL: ~10 mm; n3	9169a
	CR: 15 mm; n1	10802
Schoinobates volans	CR: 14.6 mm; n1 preserved	549
WEANING MASS		
Gymnobelideus leadbeateri	30-38 g; 120 d	10080
Petaurus australis	285 g	10081
Petaurus breviceps	4.5-5 g; 40 d	10120
	45 g; 4 mo	10081
Pseudocheirus herbertensis	115-120 g; pouch exit	4192
	200 g	10081
Pseudocheirus peregrinus	~100 g; 1st solid food	8205
	f: ~310 g; 6 mo	8204
	m: ~350 g; 6 mo	8204
	400 g	10081
Schoinobates volans	24.2 g; teat detachment	10141
	150 g; pouch exit	10141
	500-600 g; independent	10141
LITTER SIZE		
Dactylopsila trivirgata	1-2 yg; n3 lt	11159
	1.5 yg	10081
	2 yg; n1 lt	11159
Gymnobelideus leadbeateri	1.5 yg; n24 lt	10080
Petaurus australis	1 yg ea; n2 lt	4636
	1 yg ea; n4 lt	2103
	1 yg; n10 lt	3862
	mode 1; r1-2	2104
Petaurus breviceps	1.5; r1-2; n22 lt	12128
	1.57; r1-2; n14 lt	10120
	1.8 yg; 81% twins; r1-2; n116 lt	10533
Petaurus norfolcensis	1.5; r1-2; n15 lt	12128
Pseudocheirus herbertensis	1.8; r1-3; n12 lt	4192
Pseudocheirus peregrinus	1-3 ova	5083
	1 yg; n1 lt	2841
	1.7 yg; r1-3; n22 lt, 6 f	9169a
	1.94; mode 2; r1-3; n84 lt	5017
	2-3 emb	9853
	mode 2; r1-4	8205
	2 enlarged teats; r1-3; n58 f	5083, 10802
	4 emb; n1 lt	3353
Schoinobates volans	1 emb ea; n17 f	549

SEXUAL MATURITY		
Gymnobelideus leadbeateri	f: 1st repro: 1.3 yr	10080
	m: change scrotal color: 378-550 d	10080
Petaurus breviceps	f: 1st mate: 236.5 d; r227-246; n2	9669
	f: 1st birth: 15 mo; n1	10120
Pseudocheirus herbertensis	f: 1st birth: 16 mo; n2	4192
Pseudocheirus peregrinus	f: 1st conceive: 11 mo	9169a
	f: 12.5 mo; n50	8205
	m: testes descend: ~10-12 mo	8205
	m: 1st sire: 11 mo	9169a
	f/m: 12 mo	5017
Schoinobates volans	f/m: 1.5-2 yr	10141
CYCLE LENGTH		
Petaurus breviceps	rpy to estrus: 11-13 d; n3	10120
	29 d; n1	10120
Pseudocheirus peregrinus	28 d	5083
Schoinobates volans	rpy to estrus: 0 d	10998
GESTATION		
Gymnobelideus leadbeateri	<20 d	10080
Petaurus breviceps	16 d; r15-17; n3	10120
	16 d; n11 lt	9669
LACTATION		
Gymnobelideus leadbeateri	pouch exit: 87.4 d; r80-93; n5	10080
	independent: est 120 d	10080
Petaurus australis	pouch life: 100 d	4636
	wean: 5 mo	9383
Petaurus breviceps	teat attachment: 40 d	10120
	pouch exit: 74 d	9669
	wean: 4 mo	10120
Petaurus norfolcensis	wean: 4.0 mo	10081
Pseudocheirus herbertensis	pouch exit: 115-120 d	4192
	wean: est 150-160 d	4192
Pseudocheirus peregrinus	teat attachment: 42 d	5017
	pouch exit: 100-120 d	9169a
	pouch exit: ~120 d	10802
	pouch exit: 125-130 d	5017
	solid food: 4-5 mo	8205
	wean: 145-220 d	5017
	wean: 5-7 mo	8204
	nurse: 5-8 mo	8205
	wean: 160 d, (4-6 wk after pouch exit at 110-120 d)	9169
	wean: 6-7 mo	10802
Schoinobates volans	teat attachment: 6 wk	1987
	teat attachment: 2-3 mo; field data	10141
	teat attachment: 93 d	10998
	pouch exit: 170 d	10998
	pouch exit: 6 mo	4635
	wean: 7.5 mo	10081
	wean: 230 d	10998
INTERLITTER INTERVAL		
Petaurus australis	2 yr	2103
Pseudocheirus peregrinus	3-10.25 mo; n16 intervals, 6 f	9169a
LITTERS/YEAR		
Gymnobelideus leadbeateri	1-3	10080
Petaurus australis	1	4636
Petaurus breviceps	1-2	10120
Pseudocheirus herbertensis	mode 1; r1-2	4192
Pseudocheirus peregrinus	mode 1; r1-2	5083
	1-2	5017
	1-2	8205
Schoinobates volans	1	11010

SEASONALITY

Dactylopsila trivirgata	mate: Aug, perhaps Mar; few data; QLD, Australia	11159
	birth: Feb-Aug; QLD, Australia	10081
Gymnobelideus leadbeateri	birth: Mar-Dec, peaks Apr-June, Oct-Dec; VIC, Australia	10080
Petaurus australis	birth: Apr, June, Aug-Dec; few data; VIC, Australia	4993
	birth: Apr, Nov-Dec; n3; NSW, Australia	4993
	birth: June-Dec, peak July-Sept; n10; NSW, Australia	3862
	birth: yr rd, except Nov; QLD, Australia	4993
	birth: Aug-Oct; n4; VIC, Australia	2103
	birth: yr rd; QLD, Australia	9383
Petaurus breviceps	mate: June-Nov; n14; captive	10120
	mate: peak Sept-Dec; VIC, Australia	10533
	birth: Mar-Sept, Dec, peaks Apr, June; n22; zoo	12128
	birth: July; VIC, Australia	10917
	birth: Aug-Dec; Australia	10081
	birth: Sept-Nov; VIC, Australia	10533
Petaurus norfolcensis	birth: Jan-Apr, June-Oct; n15; zoo	12128
	birth: June-Jan; Australia	10081
Pseudocheirus dahli	mate: yr rd; Arnhem land, Australia	2214
Pseudocheirus herbertensis	birth: Apr-Dec, peak May-July 74%; lab	4192
	birth: June-July, Dec; Australia	10081
Pseudocheirus peregrinus	mate: May-Aug; TAS, Australia	3353
	birth: Apr-Aug, Oct-Feb; n22; captive	9169a
	birth: Apr-Dec, peak end May-early June, mid Oct; VIC, Australia	5017
	birth: Apr-Dec, peak May-June; VIC, Australia	8205
	birth: peaks May-Aug, Nov-Dec; VIC, Australia	4822
	lact f: May-Dec; VIC, Australia	5083, 10802
Schoinobates volans	spermatogenesis: Dec-July, peak Mar; NSW, Australia	534
	testes: large Mar, small Aug-Sept; NSW, Australia	534
	birth: Apr-May; NSW, Australia	10141
	birth: July-Aug; VIC, Australia	10917

Order Marsupialia, Family Macropodidae

Kangaroos and wallabies (*Aepyprymnus* 1-4 kg; *Bettongia* 1-2 kg; *Caloprymnus* .65-1.5 kg; *Dendrolagus* 6-10 kg; *Dorcopsis* 2-7 kg; *Hypsiprymnodon* 400-700 g; *Lagorchestes* 2-5 kg; *Lagostrophus* 1-2.5 kg; *Macropus* 3-90 kg; *Onychogalea* 5 kg; *Peradorcas* 1-1.5 kg; *Petrogale* 1-9 kg; *Potorous* 0.7-2 kg; *Setonix* 2-5 kg; *Thylogale* 3.5-6.5 kg; *Wallabia* 11-17 kg) are terrestrial herbivores in Australia and New Guinea. *Caloprymnus* is presumed extinct.

Female *Lagorchestes conspicillatus* have one more chromosome than males, while male *Potorous tridactylus* and *Wallabia bicolor* have one more than females (4013). *Macropus* males are often larger than females (799, 7646, 9861). A postpartum estrus (prepartum in *Wallabia*) and embryonic diapause are usual (590, 791, 1390, 1410, 1861, 2857, 3328, 3329, 3333, 3873, 3947, 4366, 4537, 4538, 4539, 5159, 5359, 5360, 5362, 5814, 6449, 6995, 7217, 7219, 7834, 8544, 8672, 9045, 9047, 9275, 9277, 9280, 9377, 9419, 9850, 9851, 9855, 9857, 9858, 9859, 9860, 9861, 9870, 10125, 10442, 10987, 10991, 11006, 11009, 11030, 11412, 11417, 11982; cf 6997, 6998, 7504, 9859).

Neonates open their eyes at 78-92 d (*Aepyprymnus* 5359; *Bettongia* 10991), 90-105 d (*Potorous* 3683, 4129, 4585), 107-120 d (*Petrogale* 5360), 115-150 d (*Macropus* 2859, 5814, 6994, 7216, 7646, 9861, 11426; *Thylogale* 9281). Females have 2-4 mammae (591, 1842, 3311, 3353, 5363, 8089, 8295, 10991).

Macropus eugenii is the "laboratory rat" of macropod reproductive physiology as its extensive literature attests (see 11008 for pre-1987 review). Recent descriptions of endocrinology (1665, 3330, 4804, 4812, 5431, 9867, 10395a, 11005), sex differentiation (7990, 9863, 11004), sperm and testicular maturation (5117, 5428, 5431, 6941, 9793a, 11732), milk and lactation (2049, 2705, 7245, 7248, 7879) are available.

For other macropods, details of anatomy (3352, 3359, 4411a, 6968, 7504, 8291, 8293, 8296, 9208, 9851, 9856, 10989, 11413), endocrinology (1200, 1537, 1618, 1619, 1857, 2184, 3169, 3170, 6454, 9279, 9790, 9860, 11001a, 11412), sperm (5074, 7871, 9209, 9210, 9746, 9793a), gestation (3357, 9861), placentation (1200a, 1201, 1549, 9853), birth (718, 1588, 3299, 3355, 4437, 9858, 10986), marsupium (1745, 1945, 6080, 11983),

lactation (2184, 6449, 9857), milk composition (497, 1066, 1945, 1985, 2160, 4026, 4070, 5445, 5499, 5500, 6374, 6375, 7215, 7247b, 7249, 8674, 10162, 11982a), weaning (9276), and seasonal quiescence (2176, 2177, 2183, 6590, 6592, 6593) are available.

NEONATAL MASS		
Bettongia gaimardi	307 mg; sd15; n5	9278
Bettongia lesueuri	317 mg; r267-344; n9	10991
Macropus agilis	634 mg; r580-660; n5	7216
Macropus eugenii	413.8 mg; r363-492; n4	11009
	444.2 mg; sd61; n25	11451
Macropus fuliginosus	934.33 mg; r828-1113; n6	8666
Macropus giganteus	898.67 mg; r740-1000; n3	8666
Macropus parma	506.7 mg; se10; r436-569; n17	6994, 6999
Macropus rufogriseus	537 mg; r453-690; n4	7219
	558 mg; r493-600; n6	11411
	700 mg; n6	6597
Macropus rufus	817 mg; r745-900; n5	9861, 9862
Petrogale xanthopus	508 mg; r460-572; n3	8672
Potorous tridactylus	333 mg; n1	5072
Setonix brachyurus	310-350 mg	11981
	382.5 mg; se43.6; r309.4-477.4; n22	9910
	500 mg	591
Thylogale billardieri	420 mg; n3 yg	9281
NEONATAL SIZE		
Bettongia lesueuri	CR: 13.75 mm; r13.00-14.30; n9	10991
Macropus fuliginosus	CR: 22.1 mm; n1; near term	8671
Macropus parma	CR: 18.33 mm; se0.21; r16.5-19.7; n17	6994, 6999
Macropus rufogriseus	CR: 16.6 mm; r15.8-17.5; n6	11411
Potorous tridactylus	CR: 14.7-16.1 mm	4129
	CR: 18 mm; r18-19; n4	5072
WEANING MASS		
Bettongia gaimardi	354.6 g; sd50; r291-460; n17; pouch exit	9277
	1046-1400 g; 23-24 wk	9278
Bettongia penicillata	min 145 g; pouch exit	10125
Macropus agilis	2286.7 g; sd256.8; n12; pouch exit	7216
Macropus parma	754 g; se25; r621-868; n10; 210 d, pouch exit	6994, 6999
Macropus rufus	~5 kg; pouch exit	9862
Peradorcas concinna	300 g	7834
Petrogale penicillata	735 g; sd26; r705-750; n3; pouch exit	1909
Potorous tridactylus	90-145 g; 1st pouch exit	3683
Setonix brachyurus	mode ~500 g; pouch exit	9910
Thylogale billardieri	1029 g; sd112; r857-1170; n6; pouch exit	9281
WEANING SIZE		
Macropus parma	276.9 mm; se2.9; r264-291; n10; 210 d, pouch exit	6994, 6999
LITTER SIZE		
Aepyprymnus rufescens	1 yg ea; n27 lt	5359
	1.07; r1-2; n15 lt	12128
Bettongia gaimardi	1 emb	3357
	1 yg ea; n23 lt	9278
	1 emb ea; n70 f	3359
	1.04; r1-2; n26 lt	12128
Bettongia lesueuri	1 ea; n8 lt	10991
	1 yg; n1	9194
Bettongia penicillata	1 yg; n~10 lt	11310
Caloprymnus campestris	1 yg; n7 f	3257
Dendrolagus matschiei	1 yg	8089
	1 yg; n6 lt	5111a
	1 ea; n16 lt	2152
Dendrolagus ursinus	1; n1 lt	12128
Dorcopsis vanheurni	1 yg	7983
Hypsiprymnodon moschatus	2; n1 lt	8901
	2 yg	5363
Lagorchestes conspicillatus	1 yg	5362

Lagostrophus fasciatus	1 yg ea; n9 f	10989
	mode 1 cl; r1-2	10989
Macropus agilis	1 yg ea; n15 lt	7763
	1 yg ea; n64 lt	7216
	1.01 yg; r1-2; n134 lt	5814
Macropus dorsalis	1 ea; n3 lt	12128
Macropus eugenii	1.01 yg; r1-2; n262 births	5154
	1.07 yg; r1-2; n14 lt	12128
Macropus fuliginosus	1.00; r1-2; n280 lt	7955
	1.01 yg; r1-2; n~170 lt	8666
Macropus giganteus	1.0 yg ea; n~118 lt	8666
	1.00 yg; r1-2; n201 lt	8286
	1.02 yg; r1-2; n41 lt	12128
Macropus irma	1 yg	9120
Macropus parma	1 cl; 1 yg	6997
Macropus parryi	1 yg ea; n16 lt	6996
Macropus robustus	1 yg; 0.2% twins	2859
Macropus rufogriseus	1 yg ea; n9 lt	9426
	1.01 yg; r1-2; n103 lt	3311
	1.16 yg; r1-3; n51 lt	12128
Macropus rufus	1 yg ea; n44 lt	12128
	1 yg; 1/1000 twins	9859
Onychogalea fraenata	1 yg; n1 lt	12128
Petrogale penicillata	1 ea; n7 lt	12128
	1 yg ea; n~12 lt	5360
Petrogale xanthopus	1.06 yg; r1-2; n36 lt	12128
Potorous longipes	1 yg; n1	9720
Potorous tridactylus	1 yg ea; n5 f	4585
	1 yg ea; n30+ lt	5072
	1 yg ea; n38 lt, 13 f	648
	1 yg ea; n42 lt, 21 f	4129
Setonix brachyurus	1 yg; n1	3299
	1 yg ea; n23 lt	9911
Thylogale billardieri	1 yg	10474
Wallabia bicolor	1.1 yg; r1-2; n7 lt	12128
SEXUAL MATURITY		
Aepyprymnus rufescens	f: 1st mate: 313 d; n1	5359
	m: 1st mate: 379 d; n1	5359
Bettongia gaimardi	f: 1st birth: 272 d; ±53	9277
	m: testosterone surges: ~30 wk; 1.5 kg	9279
	m: 1st mate or sperm in urine: 272 d; ±53	9277
Bettongia lesueuri	f: 1st mate: 217 d; r207-227; n2	10991
	f: 1st birth: 200 d	10991
Bettongia penicillata	f/m: end 1st yr	10809
Hypsiprymnodon moschatus	f/m: 1+ yr	5363
Lagorchestes conspicillatus	f/m: ~1 yr	5362
Lagostrophus fasciatus	f/m: est 1 yr	10989
Peradorcas concinna	f: 1st birth: 470 d; n1	7834
Macropus agilis	f: 1st estrus: 330-480 d	5814
	m: epididymal sperm: 385-401 d	5814
Macropus eugenii	f: 1st mate: < 1 yr	301, 590
	m: rapid testes growth: 18-20 mo	5155
	m: 2 yr	590
Macropus fuliginosus	f: teat eversion: 657.4 d; r426-949; n9	8671
	f: 1st mate: 647.1 d; r493-791; n9	8671
	f: 1st birth: 749.1 d; r524-1096; n11	8671
	f: 1st birth: 20 mo	8667
	m: tubular sperm: 20 mo	8667
	m: active sperm: 31 mo	8671
Macropus giganteus	f: teat eversion: 636 d; r543-751; n8	8671
	f: 1st mate: 639.3 d; r579-707; n3	8671
	f: 1st mate: mode 21 mo; r17-28	5810
	f: 1st birth: 812.6 d; r616-1446; n9	8671
	f: 1st birth: 20-36 mo	8669

	m: epididymal sperm: 20-30 mo	5810
	m: active sperm: 48 mo	8671
	m: 1st sperm: 20-36 mo	8669
	f/m: 1st repro: 16 mo	8286
Macropus parma	f: 1st mate: 357-491 d	6997, 6999
	f: 1st mate: est 19 mo	7001
	m: 1st sperm: 19-20 mo	6997
	m: 1st sire: 24-25 mo	6997
Macropus parryi	f: 1st mate: 24-25 mo	5581a
Macropus robustus	f: 1st mate: 2 yr	2859
Macropus rufogriseus	f: 1st mate: 13 mo	7054
	f: 1st lact: 14-18 mo	1665
	m: testes size: 18-22 mo	1665
Macropus rufus	f: 1st mate: min 14-20 mo	3525, 9858, 9861
	m: 1st sperm: 2-3 yr	3525, 9858, 9861
Petrogale penicillata	f: 1st mate: 540 d	5360
	m: 1st mate: 590 d	5360
Petrogale xanthopus	f: 1st conceive: ~16 mo	4993
	f: 1st birth: 541.3 d; sd112.6; r321-675; n10	8672
Potorous tridactylus	f: 1st birth: 11.5-17 mo	9718
	m: 600 g	5074
	f: 1st mate: ~1 yr	4128
Setonix brachyurus	f: 1st mate: 16-23 mo; ~2.5 kg	2811
	f: 1st birth: 8-10 mo; n5	9910
	m: 1st sire: 389-455 d; n7 d	9910
Thylogale billardieri	f: 1st birth: 14-15 mo	9280
	m: 14 mo; 3.9-4.75 kg	9280
Wallabia bicolor	f/m: 15 mo	5813
CYCLE LENGTH		
Aepyprymnus rufescens	rpy to estrus: 8.5 d; sd4.2; r4-14; n4	7504
	r21.0-25.0 d; n5 cycles, 4 f	5359
	33.8 d; sd2.1; mode 33-36; r28-36; n11 cycles, 4 f	7504
Bettongia gaimardi	rpy to estrus: 18.9 d; sd1.3; n16	9277
	22.6 d; sd3.5; r17-37; n46 cycles, 10 f	9277
Bettongia lesueuri	rpy to estrus: 21.50 d; mode 22; r16-28; n14	10991
	22.17 d; mode 23; r11-35; n18 cycles	10991
Macropus agilis	r29-34 d; n8 cycles, 4 f	5814
	32.4 d; sd2.2; r28-41; n23 cycles, 6 f	7216
Macropus eugenii	estrus: 12 h; n7 cycles, 7 f	6373
	rpy to estrus: 26.4 d; sd0.57; r25.5-28.5; n10 cycles with gestation, 10 f	7217
	rpy to estrus: 30.4 d; sd0.99; r28.5-31.5; n10 cycles, 10 f	7217
	28.4 d; se0.1	799
	29.4 d; sd1.26; r28-30; n10 cycles with gestation, 10 f	7217
	29.6 d; se0.45; r27-32; n12 cycles with gestation, 6 f	8674a
	30.6 d; sd1.17; r29-33; n10 cycles, 10 f	7217
	32.1 d; se0.58; r30-37; n16 cycles, 6 f	8674a
Macropus fuliginosus	rpy to estrus: 6.3 d; sd0.1	8667
	34.6 d; sd3.2; r25-48; n165 cycles, 8 f	8667
	34.85 d; sd4.42; r25-55; n292 cycles, 31 f	8671
Macropus giganteus	r41-43 d; n4 f	5810
	45.58 d; sd9.82; r26-81; n105 cycles, 19 f	8671
Macropus parma	rpy to estrus: 6-15 d	6997
	40.34 d; ±0.47; r36-48; n51 cycles	6995, 6998
	41.78 d; se0.72; r36-59; n58 cycles, 16 f	6997
Macropus parryi	rpy to estrus: 6 d; n1	5581a
	42.2 d; r41-44; n5	5581a
Macropus rufogriseus	rpy to estrus: 26 d	11459
	32.9 d; sd2.3; r28-40; n38 cycles, 8 f	7219
	33.4 d; sd2.3; r30-38; n14 cycles, 4 f	7219
Macropus rufus	rpy to estrus: 34.0 d; se0.89; n8	9855
	rpy to estrus: 34.46 d; sd1.92; n13 cycles, 10 f	9857
	34.30 d; se0.50; n23 cycles, 9 f	9858
	34.64 d; sd2.22; n42 cycles, 13 f	9857

	34.81 d; se0.63; n16 nonpreg	9855
	35.53 d; se0.52; n17 preg	9855
Peradorcas concinna	33.73 d; sd1.65; mode 35; r31-36; n52 cycles	7834
Petrogale penicillata	rpy to estrus: 31 d; n2	5360
	31.06 d; r30.2-32.0; n5 cycles, 5 f	5360
Petrogale xanthopus	rpy to estrus: 34.375 d; n1	8672
	r32-37 d; n3	8672
Potorous tridactylus	41.5 d; r39-44; n5 cycles	5072
Setonix brachyurus	luteal phase: 20 d	9851
	rpy to estrus: 26 d	9859
	28 d	9851, 9859
Thylogale billardieri	rpy to estrus: 29.2 d; r27-31; n6	9280
	remove cl to estrus: 11 d; n2	9280
	30.3 d; r30-32; n12 cycles	9280
Wallabia bicolor	rpy to estrus: 26 d; r23-28; n3	9859
	32 d; r27-44; n5	9859
GESTATION		
Aepyprymnus rufescens	rpy to birth: 18.7 d; r18-20.8	5359
	r22.0-23.6 d; n5	5359
	25.5 d; sd9.8; r21-30; n2	7504
Bettongia gaimardi	rpy to birth: 17.6 d; sd0.9; r16-19	9277
	19.5 d	9746
	21.3 d; sd0.8; 20-22; n15	9275, 9277
Bettongia lesueuri	rpy to implantation: 21-22 d	11459
	rpy to birth: 20.00 d; mode 20; r17-24; n14	10991
	21.33 d; mode 21; r22-23; n6	10991
	21-21.5 d; n1	10442
Bettongia penicillata	rpy to birth: 17.75 d; sd1.12; n20	10125
Dendrolagus matschiei	~32 d	8089
Lagorchestes conspicillatus	rpy to birth: 28-30 d	5362
	29-31 d; no lactation	5362
	5 mo with diapause	5362
Macropus agilis	rpy to birth: 25.85 d; r24.2-27.7; n4	5814
	29.4 d; sd0.9; r28-31; n16 lt, 8 f	7216
	30.52 d; r29.0-33.1; n4 lt, 4 f	5814
Macropus eugenii	rpy to birth: 26.2 d; se0.67; r23.5-26.5; n10	7217
	rpy to birth: 26.9 d; sd0.56; n10	12041
	rpy to birth: 27.30 d; sd0.95; n30	11451
	rpy to birth: 27.6 d; sd1.2; n16	11009
	28.4 d; se0.2; n12	791, 799
	12 mo with diapause	791, 799
Macropus fuliginosus	30.56 d; sd2.55; mode 30; r27-46; n115 lt, 30 f	8666
	30.8 d; sd1.8; r28-35; n25 lt, 6 f	8667
Macropus giganteus	36.44 d; sd1.63; mode 36; r33-45; n53 lt, 23 f	8666
	36.7 d; r35-38; n14	5810
Macropus parma	rpy to birth: 31.3 d; r30.5-32; n6	6998
	34.54 d; se0.13; mode 34-35; r33-36; n28 lt, 16 f	6995, 6997
Macropus parryi	36.3 d; r34-38; n3	5581a
Macropus robustus	rpy to birth: 33-34 d	5812
	35 d; mating to birth; 8-9 mo with diapause	5812
Macropus rufogriseus	rpy to birth: 22.0 d; sd4.5	7219
	rpy to birth: 26.8 d; sd0.6; n6	11412
	rpy to birth: 27 d; se0.2; n5	2183
	rpy to birth: 27.3 d; r26-28; n3	3311
	mating to ppe: 30 d; sd1.4; r28.5-34; n12 lt, 5 f	7219
	mating to ppe: 30.5 d; sd1.7; r29-41; n19 lt, 8 f	7219
	max 12 mo with diapause	3311
Macropus rufus	rpy to birth: 31.34 d; se0.35; n8	9855
	rpy to birth: 31.64 d; sd0.65; n7 f	9857
	33.00 d; sd0.32; n20 lt, 14 f	9857
	33.17 d; se0.16; n14	9855
Peradorcas concinna	30 d; n1	7834
Petrogale penicillata	rpy to birth: 29.1 d; r27.6-30; n6	5360
	30.8 d; r30-32; n5	5360

	199.5 d; r193-206; n4 with diapause	5360
Petrogale xanthopus	rpy to birth: 31.5 d; n1	8672
	31.8 d; sd0.76; 31-33 d; n5 birth, 3 f	8672
Potorous tridactylus	rpy to birth: 29 d; n4	9870
	30, 32-43, 34, 37.5 d; n4	5072
Setonix brachyurus	rpy to birth: mode 25 d; r24-27; n26	9910
	rpy to birth: 25.8 d; r25-26; n17	9911
	25-26 d; n1	9910
	27 d	9851, 9853, 9859
Thylogale billardieri	rpy to birth: 28.7 d; r27-31; n6	9280
	30.2 d; mode 30; r30-31; n5	9280
Wallabia bicolor	35 d; r33-38; n4 lt, 2 f	9859
	36.8 d; r35-38	7983
LACTATION		
Aepyprymnus rufescens	teat attachment: 87.3 d; r84-92	5359
	pouch exit: 114 d; r105-119; n5	5359
	nurse: ~164 d	5359
Bettongia gaimardi	teat attachment: ~5 wk	9278
	pouch exit: 106.1 d; sd2.5; r101-111	9275, 9277, 9278
	wean: 40-60 d after pouch exit	10474
	wean: 8-9 wk after pouch exit	9278
Bettongia lesueuri	pouch exit: 116.25 d; mode 115; r113-122; n4	10991
	nurse: 23-74 d after pouch exit	10991
Bettongia penicillata	pouch exit: 97.9 d; sd3.28; n25	10125
Dendrolagus matschiei	solid food: 266 d	8089
	pouch exit: 305 d	8089
Hypsiprymnodon moschatus	pouch exit: 21 wk	5363
Lagorchestes conspicillatus	pouch exit: 5 mo	5362
Macropus agilis	pouch exit: 208.5 d; r200-220; n9	5814
	pouch exit: 219.3 d; sd9.7; r207-235; n15	7216
	nurse: 321.5 d; r288-340; n6	7216
Macropus eugenii	teat attachment: 100-110 d	7646
	solid food: 190-220 d	7646
	pouch exit: ~6 mo	590
	pouch exit: 241 d; se8; r222-259; n4	4811
	pouch exit: 35-40 wk	11699
	pouch exit: 250 d	799
	wean: 269 d; se4; r258-282; n6	4811
	wean: 45 wk	4029b
Macropus fuliginosus	1st pouch exit: 298.40 d; sd34.29; r236-359; n25	8666
	pouch exit: 300 d	8667
	permanent pouch exit; 323.12 d; sd22.92; r280-359; n24	8666
Macropus giganteus	pouch exit: 297 d; r277-308; n12	5810
	1st pouch exit: 283.86 d; sd24.70; r247-329; n21	8666
	permanent pouch exit: 319.23 d; sd18.36; r297-359; n22	8666
	nurse: 18 mo	5810
Macropus parma	teat attachment: ~110 d	6994
	1st pouch exit: 160-175 d	6997
	permanent pouch exit: 211.9 d; se1.0; r207-218; n10	6997
	wean: 2.5-3.5 mo after pouch exit	6997
Macropus parryi	pouch exit: 9 mo	5581a
	nurse: 10-15 mo	5581a
Macropus robustus	teat attachment: 120 d	2859
	pouch exit: 7-8 mo	2857, 2859
	wean: 13 mo	2859
Macropus rufogriseus	pouch exit: 77-204 d	3311
	pouch exit: 280 d	7219
	pouch exit: 9-10 mo	5335
	wean: 4-5 mo after pouch exit	5335
Macropus rufus	pouch exit: 236.50 d; se2.66; n10	9855
	wean: 360 d	9379
Peradorcas concinna	pouch exit: 160 d	7834
	independent: 175 d	7834, 9485

Petrogale penicillata	pouch exit: 189-227 d	5360
	pouch exit: 194 d; sd8; r188-205; n4	1909
	independent: 59-86 d after pouch exit	1909
Petrogale xanthopus	pouch exit: 194.9 d; sd3.9; 190-201; n7	8672
Potorous tridactylus	teat attachment: 64 d; n1	5072
	1st pouch exit: 103.2 d; se2.0; r100-106; n6	3683
	pouch exit: 17 wk	4585
	permanent pouch exit: 119.7 d; se1.6; r115-124; n6	3683
	pouch exit: 140 d	4133
	wean: 21-23 wk	4585
Setonix brachyurus	pouch exit: 185-195 d; r177-196; n19	9910
Thylogale billardieri	pouch exit: 202 d; sd7; r196-212; n6	9281
	nurse: 3 mo after pouch exit	9281
Wallabia bicolor	pouch exit: 256 d	9859
INTERLITTER INTERVAL		
Lagorchestes conspicillatus	5 mo	5362
Macropus agilis	potentially 7-8 mo	5814
Macropus eugenii	1 yr with diapause	791
Macropus fuliginosus	372.02 d; sd61.39; r299-559; n45 intervals, 24 f	8666
	398.2 d; sd82.9; r278-559	8667
Macropus giganteus	306 d; r299-312; n4 with diapause	5810
	356 d; r338-368; n5	5810
	362.59 d; sd56.90; r283-523; n27 intervals, 15 f	8666
Macropus robustus	8-9 mo	5812
Macropus rufogriseus	9 mo	7054
Petrogale penicillata	~200 d	5360
Potorous tridactylus	122.8 d; se2.2; r118-129; n5	3683
	min 5-6 mo; yg lives	4585
Wallabia bicolor	~8 mo	7983
SEASONALITY		
Aepyprymnus rufescens	mate: est yr rd; QLD, Australia	7504
	birth: yr rd; few data; captive	5359
	birth: yr rd; n15; zoo	12128
Bettongia gaimardi	estrus: fall-spring; captive	9746
	mate: winter; TAS, Australia	3353
	birth: none Jan, Sept; n26; zoo	12128
	birth: yr rd; captive	9275
Bettongia lesueuri	mate: yr rd, peaks Apr-May, Nov-Dec; captive	10991
Bettongia penicillata	birth: yr rd; captive	10125
Caloprymnus campestris	pouch yg: June, Aug, Dec; s central Australia	3257
Hypsiprymnodon moschatus	mate: Feb-May, rainy season; few data; NSW, Australia	8901
	conceive: Feb-June; QLD, Australia	5363
Lagorchestes conspicillatus	birth: yr rd; QLD, Australia	5362
	birth: yr rd, peaks Mar, Sept; Barrow I, WA, Austraila	10474
Lagostrophus fasciatus	birth: peak est before July; WA, Australia	5071, 10989
Macropus agilis	birth: yr rd; n135; QLD, Australia	5814
	birth: yr rd; captive	7216
Macropus eugenii	spermatogenesis: yr rd; captive	4539
	prostate: peak Jan; captive	5155
	mate: Jan-Feb; captive	1664
	mate: end Jan; SA, Australia	3333
	mate: Feb-May; captive	9047
	ovulate: Dec-Jan; SA, Australia	3333
	obligate diapause: June-Dec; captive	11003
	implant: end Dec; SA, Australia	3333
	birth: Jan-Aug, most Jan-Feb; SA, Australia	301
	birth: end Jan-early May; n10; captive	4811
	birth: Apr-Oct; n14; zoo	12128
Macropus fuliginosus	birth: yr rd; captive	8671
	birth: yr rd, peak summer; SA, Australia	8667
	birth: yr rd, peak Nov; VIC, Australia	7956
Macropus giganteus	epididymal sperm: yr rd; s QLD, Australia	5810
	mate: yr rd, peak Oct-Dec; s QLD, Australia	5810
	birth: yr rd; captive	8666, 8671

	birth: yr rd, 70% Dec-Feb; TAS, Australia	8286
	birth: yr rd, low winter; e Australia	8669
Macropus parma	mate: yr rd, low July-Dec; captive	6998
	birth: peak Feb-June; NSW, Australia	7000
	birth: yr rd; n1 yr; Mar-July; n1 yr; New Zealand	7001
Macropus parryi	birth: yr rd, peak Mar-July; NSW, Australia	5581a
Macropus robustus	mate: yr yr, peak Sept-Nov; WA, Australia	2857
	birth: yr rd; WA, Australia	9421
Macropus rufogriseus	prostate: max Feb-Mar, low Aug-Oct; New Zealand	1665
	mate: peak end summer; TAS, Australia	4128
	birth: Jan-July; TAS, Australia	7219
	birth: Jan-Aug; se Australia	7219
	birth: Feb-Mar; New Zealand	1665
	birth: July-Dec, peak Aug; captive	3311
	birth: yr rd, peak Jan; QLD, Australia	7054
Macropus rufus	estrus: yr rd; captive	9858
	birth: yr rd; NSW Australia	3525
	birth: yr rd; WA, Australia	9421
	birth: yr rd; n44; zoo	12128
Peradorcas concinna	small pouch young: Aug; NT, Australia	9485
Petrogale penicillata	birth: yr rd; captive	5360
Petrogale xanthopus	mate: yr rd; captive	8672
	birth: yr rd; n36; zoo	12128
Potorous longipes	birth: est yr rd; se Australia	10474
Potorous tridactylus	mate: winter; TAS, Australia	3353
	mate: yr rd, peak Dec-Feb; TAS, Australia	6733
	mate: yr rd, peaks end Aug-early Oct, Jan; TAS, Australia	4127
	mate: est yr rd; TAS, Australia	4128
	birth: not May-June; TAS, Australia	5072
	birth: yr rd; TAS, Australia	4585
	birth: yr rd, peak Sept-Nov; captive	9718
	birth: yr rd; captive	4133
	birth: yr rd; captive	648
Setonix brachyurus	anestrus: Aug-Jan; WA, Australia	9849
	mate: yr rd; WA, Australia	9912
	birth: Jan-June; Rottnest I, Australia	9912
	birth: yr rd; mainland, Australia	9909, 9912
	birth: peak Feb-Mar; WA, Australia	9849
	birth: peak Mar; WA, Australia	9910
	birth: yr rd, peak Feb-May; WA, Australia	2811
Thylogale billardieri	mate: yr rd; TAS, Australia	9280
	birth: peak Apr-June; TAS, Australia	9280
Wallabia bicolor	birth: yr rd; n18; captive	12128
	birth: yr rd; QLD, Australia	5813

Order Marsupialia, Family Phascolarctidae, Genus *Phascolarctos*

Koalas (*Phascolarctos cinereus* 4-14 kg) are solitary, arboreal folivores of Australian forests. Males are larger than females (7107, 10081). Their Sertoli cells have crystalloid inclusions (4370a, 5696) and their sperm are structurally heterogeneous (11681). A copulatory plug is produced (10122). Females have 2 teats and a posteriorly opening pouch (10698, 11901). Young are coprophagous (7358). Two symposium volumes review koala biology (807, 6345). Details of semen (11681) and placentation (1549, 8294) are available.

NEONATAL MASS		
Phascolarctos cinereus	360 mg	10917
WEANING MASS		
Phascolarctos cinereus	0.82-1.49 kg; n14; 12 mo	10122
LITTER SIZE		
Phascolarctos cinereus	1 yg; n4 lt	3300
	1; 2 occasional	10698
	1 yg ea; n31 lt, 25 f	10122

SEXUAL MATURITY		
Phascolarctos cinereus	f: 1st conceive: 2 yr; n3	10122
	f: 12-14 mo	8706
	f: end 2 yr	2884
	f: end 2 yr	6894
	m: 1st sire: 3 yr	10122
CYCLE LENGTH		
Phascolarctos cinereus	27-30 d	2884, 10122
GESTATION		
Phascolarctos cinereus	25-30 d	8700, 8706
	34.6 d; r34-36; n7 lt	10122
LACTATION		
Phascolarctos cinereus	teat attach: ~5.5 mo	8700
	pouch exit: 5 mo; n1	3300
	pouch exit: ~5-6 mo	7107
	pouch exit: 6-7 mo	2884
	pouch exit: 209-224 d	10122
	pouch exit: 240-270 d	9379
	wean: ~10 mo	8700, 8704, 8706
	wean: 11 mo	6898
	wean: ≤12 mo	2884
	wean: 12 mo	10122
	independent: 1 yr	7107
INTERLITTER INTERVAL		
Phascolarctos cinereus	9-12 mo	2884
	1 yr	7107
	every other yr	10917
SEASONALITY		
Phascolarctos cinereus	mate: ~Oct; se Australia	1550
	birth: Jan-Mar; VIC, Australia	6898
	birth: Feb-Sept; VIC, Australia	7107
	birth: Oct-Jan 80%; QLD, Australia	10122
	birth: Nov-Mar; QLD, Australia	6899

Order Marsupialia, Family Vombatidae

Wombats (*Lasiorhinus krefftii*, *L. latifrons*, *Vombatus ursinus*) are 20-40 kg of muscle inside a tough integument. These crepuscular and nocturnal, solitary herbivores burrow extensively in the semi-arid regions of Australia. Females are larger than males (5336). A copulatory plug occurs (2158). Details of anatomy (1329, 8000), endocrinology (8400, 9790), and placentation (8294) are available.

NEONATAL MASS		
Lasiorhinus latifrons	500 mg; n3	3632
NEONATAL SIZE		
Lasiorhinus latifrons	2 cm; n3	3632
WEANING MASS		
Lasiorhinus latifrons	1.4-3.4 kg; 8 mo	3632
	2 kg; solid food	11587
	5.5-7.5 kg; 1 yr	3632
	13.55 kg; ±0.79; n10; 1st capture	11587
Vombatus ursinus	5.7 kg; n2	8747
LITTER SIZE		
Lasiorhinus latifrons	1 yg; n4 lt	2158
	1 yg; n8 f	11587
Vombatus ursinus	1 yg; n1 lt	2021
	1; twins recorded	10698
SEXUAL MATURITY		
Lasiorhinus latifrons	f: 18 mo	3632
	f: lact: 18.75 kg	11587
	m: spermatogenesis: 18 mo	3632
Vombatus ursinus	f/m: 2 yr	10698

CYCLE LENGTH		
Vombatus ursinus	33 d; r32-34; n8 cycles, 4 f	8400
GESTATION		
Lasiorhinus latifrons	20-21, 21-22 d; n2	2158
LACTATION		
Lasiorhinus latifrons	solid food: 265 d; n1	2158
	pouch exit: 6 mo	11587
	pouch exit: 9 mo	3632
	wean: 8-9 mo	11587
	wean: 12 mo	3632
Vombatus ursinus	pouch exit: 9.5 mo	8747
	independent: 12-15 mo	10698
SEASONALITY		
Lasiorhinus latifrons	mate: summer; SA, Australia	9852
	birth: Oct-Dec; SA, Australia	11587
Vombatus ursinus	mate: yr rd, peak summer-fall; Australia	10698
	birth: June; s VIC, Australia	8747

Order Marsupialia, Family Tarsipedidae, Genus *Tarsipes*

Honey possums (*Tarsipes spenserae*) are small (10-20 g), gregarious, arboreal inhabitants of southwest Australian heath where they subsist on pollen and nectar. Females are larger than males (2168, 9046, 10081, 11911) and have 4 teats (2373, 9046). A postpartum estrus and embryonic diapause are present (9037). The penis is not bifid (11926). Neonates open their eyes at 56 d (9046). Details of anatomy (2168, 2373, 9305) and sperm structure (4370, 4371, 10735) are available.

NEONATAL MASS		
Tarsipes spenserae	4.3 mg; r3-6; n3	9046
WEANING MASS		
Tarsipes spenserae	2 g; pouch exit	11911
	2.5 g; n2; pouch exit	2804
	4-5 g; wean	2804, 11911
LITTER SIZE		
Tarsipes spenserae	2.47 yg; mode 2 55%; r1-4; n154 lt	2804, 11911
	2.67 yg; n15 f	9037
	2.8 yg; r1-4; n20 lt	9526
	5 emb	9039
SEXUAL MATURITY		
Tarsipes spenserae	f: 1st lact: 8-10 g; n46	2804
	f: 5-6 mo, 8-10 g	11911
	m: 5-6 mo, 7-9 g	11911
GESTATION		
Tarsipes spenserae	21-28 d	7983
	~3 mo	11911
LACTATION		
Tarsipes spenserae	pouch exit: 63-69 d	9380
	wean: 90 d; n1 lt	9380
LITTERS/YEAR		
Tarsipes spenserae	est 3	9046
SEASONALITY		
Tarsipes spenserae	mate: yr rd; s WA, Australia	11911
	birth: yr rd; WA, Australia	9526
	birth: yr rd, peak spring, low Dec-Jan; WA, Australia	9046
	lact f: peaks Feb-Mar, June-July, Sept-Oct, low Dec; s WA, Australia	11911